



华虹宏力
HHGRACE

HUA HONG SEMICONDUCTOR LIMITED

華虹半導體有限公司

(Incorporated in Hong Kong with limited liability)

(於香港註冊成立之有限公司)

(SEHK:01347)

(香港聯交所股份代號:01347)

(SSE STAR MARKET:688347)

(上交所科创板證券代碼:688347)

2024 年度 環境、社會及管治報告

ENVIRONMENTAL, SOCIAL AND GOVERNANCE (ESG) REPORT



2024 ENVIRONMENTAL, SOCIAL AND GOVERNANCE REPORT

Report Preparation Instructions

This report is the ninth Environmental, Social and Governance Report of Hua Hong Semiconductor Limited, disclosing to stakeholders the Company's operating philosophy, established management practices, implemented work and results achieved in ESG topics during its operations.

Scope of the Report

The scope of this report covers Hua Hong Semiconductor Limited ("Hua Hong Semiconductor", "the Company" or "we"). Unless otherwise specified, its scope is consistent with the scope of the consolidated financial statements for the same period of Hua Hong Semiconductor (stock code: 01347.HK/688347.SH).

The factories of Hua Hong Semiconductor involved in production and manufacturing include the Shanghai Production base and Wuxi Production Base. The Shanghai Production Base covers Fab 1, Fab 2 and Fab 3, while the Wuxi Production Base covers Fab 7. Wuxi Fab 9 was completed and put into production in December 2024 and is not included in the scope of this report.

Reporting Period

This Reporting Period is from 1 January 2024 to 31 December 2024. Unless otherwise stated, the data in this report refers to the data within this period.

Basis of Preparation

This report is prepared in accordance with Appendix C2 Environmental, Social and Governance Reporting Guide (effective from 31 December 2023) of the Listing Rules issued by the Hong Kong Stock Exchange, Self-Regulatory Guidelines No.14 for Companies Listed on Shanghai Stock Exchange—Sustainability Report (For Trial Implementation) (April 2024) issued by the Shanghai Stock Exchange and the Environmental, Social and Governance (ESG) Indicator System for State-owned Listed Companies in Shanghai (Version 1.0) issued by the Shanghai Municipal State-owned Assets Supervision and Administration Commission ("Shanghai SASAC").

Data Description

The data and cases in the report come from the Company's official records of actual operations. The financial data in the report were all denominated in RMB. If there is a discrepancy between the financial data and the Company's annual financial report, the annual financial report shall prevail.

Report Acquisition Method

This report is published in electronic form. The publishing platform includes the information disclosure platform designated by the stock exchange, and can also be viewed online or downloaded from the Company's official website (<https://www.huahonggrace.com>).

Contact Us

If you have any suggestions regarding the report, please contact us through the following methods:

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Reporting Principles

Materiality

The Company has identified the material topics related to its operations that are of concern to various stakeholders as the focus of this report. While reporting on material topics in this report, attention is also paid to the characteristics of the industry and business operations in which the Company is involved. For details of the process and results of the materiality analysis of topics, please see the “Materiality Assessment” section of this report.

Accuracy

The information in this report is as accurate as possible. Among them, the measurements of quantitative information have been described with data calibrations, calculation bases and assumptions to ensure that the range of calculation errors would not be misleading to the users of the information. For details of the quantitative information and the accompanying information, please see the “ESG Data and Notes” section of this report.

Balance

The contents of this report reflect objective and truthful facts and provide unbiased disclosure of both positive and negative information concerning the Company. During the Reporting Period, no negative events were discovered that should have been disclosed but were not.

Clarity

This report is issued in simplified Chinese, traditional Chinese and English. This report contains information such as tables, model diagrams and lists of laws, regulations and relevant policies that the Company has complied with as an adjunct to the textual content of this report, so as to facilitate stakeholders to better understand the textual content of the report. To facilitate faster access to information by stakeholders, this report provides a table of contents and an index table of ESG standards.

Quantitative

This report discloses key quantitative disclosures and, where possible, historical data.

Comparability

This report maintains a consistent approach to statistics and disclosure of the same quantitative disclosure items in different reporting periods; if there are changes in the method of data collection, measurement and calculation, the relevant data will be adjusted retrospectively, and the circumstances and reasons for the adjustments will be explained in the notes to the report, so as to enable stakeholders to conduct meaningful analyses and assess the trend of the development of the Company’s ESG data levels.

Completeness

The scope of the disclosure object of this report is consistent with the scope of the Company’s consolidated financial statements.

Timeliness

This is an annual report covering the time period from 1 January 2024 to 31 December 2024

Verifiability

The cases and data in this report come from the original records of the Company’s actual operations or financial reports. The sources of data and calculation processes disclosed by the Company are traceable and can be used to support external verification checks.

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ESG Management Statement of the Board

The Company adheres to the entrepreneurial spirit of “set a course mastering the challenges ahead”, with the vision of empowering our customers through continuous innovation. We focus on “8-inch + 12-inch” and advanced “Specialty IC + Power Discrete” development strategies, providing diversified wafer foundry and related support, and strongly supporting emerging applications in new energy vehicles, green energy, and the Internet of Things. Our commitment to an exceptional quality management system ensures that we meet the stringent requirements of automotive electronic chips.

The Board places great emphasis on Environmental, Social, and Governance (ESG) matters, recognizing them as a core component of the Company’s sustainable development strategy. The Board directs the formulation of ESG management policies and development strategies, guides the identification and determination of significant ESG topics. Directors and senior executives regularly monitor and review the progress of the Company’s ESG objectives to ensure continuous improvement in environmental protection, social responsibility, and corporate governance.

During the Reporting Period, the Company reformulated its ESG management objectives and will systematically improve the full-cycle management efficiency in the fields of energy control, water recycling, greenhouse gas emissions and waste under the guidance of advanced standards, actively responding to the national “Dual Carbon” strategic orientation. Through the construction of a data-driven governance system and in-depth integration of ESG standards, we will dig deep into green economic opportunities, demonstrate strategic determination in clean technology application and industrial chain layout, and firmly determine to forge ahead on the road of sustainable development.

During the Reporting Period, the Company obtained Platinum Award rating certification from the Responsible Business Alliance, focusing on key areas such as employee rights, environmental responsibility, health and safety, business ethics, and management systems. We have systematically integrated social responsibility into production operations to ensure green, healthy, and sustainable development. The management practices and progress of ESG topics in this report were considered and approved at the meeting of the Board of Directors of the Company.

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1. Materiality Assessment

1.1 Double Materiality Analysis

In compliance with the Hong Kong Stock Exchange's Environmental, Social and Governance Reporting Guide and the Self-Regulatory Guidelines No.14 for Companies Listed on Shanghai Stock Exchange—Sustainability Report (For Trial Implementation) (the "SSE Guidelines") and with reference to international sustainability disclosure standards such as the GRI Sustainability Reporting Standards from the Global Sustainability Standards Board, the Company analyzes and evaluates each ESG topic from the dual perspectives of impact materiality and financial materiality to determine the material topics.

Hua Hong Semiconductor 2024 Topics Double Materiality Analysis Process

Step	Details
Step 1 Understand the Context of Company Activities and Business Relationships	- Gain insights into the Company's strategic plans. - Analyze the external market environment in which the Company operates. - Identify value chain segments that generate impacts and the key stakeholders.
Step 2 Conduct Stakeholder Engagement, Topic Identification and Due Diligence	- Communicate with key stakeholders to identify and determine topics related to the Company, taking into account the industry background and the Company's practices. - Conduct due diligence through engagement, surveys, interviews, and other methods to engage stakeholders and analyze and identify the impacts, risks, and opportunities of the topics.
Step 3 Materiality Assessment	- Establish key indicators and evaluation criteria using a scientific and effective methodology. - Engage senior management and financial leaders to assess the financial materiality of the topics, while involving employees, suppliers, customers, and other stakeholders to evaluate the impact materiality of the topics, thereby determining the ranking of topics based on their overall materiality.
Step 4 Topic Identification and Disclosure	After review and confirmation by the Company's Board of Directors, prioritize the disclosure of topics deemed highly material for the 2024 in this report.

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1.2 Stakeholder Engagement and Due Diligence

Based on the business and operational characteristics of our Company and leveraging the experience and practices in industry at home and abroad, Hua Hong Semiconductor has determined the following to be our stakeholders: our shareholders and senior management, customers, employees, government and regulatory authorities, partners, communities, and the public. The Company has established a stakeholder engagement mechanism to maintain close engagement with these stakeholders via various channels and methods, including websites, media, meetings, reports, and social activities. This mechanism enables the Company to promptly address the concerns of stakeholders.

Hua Hong Semiconductor Stakeholder Concerns and Engagement Methods

Key Stakeholders	Description	Concerned Topics	Engagement and Response Methods
Shareholders and senior management	Domestic and foreign investors holding equity and debt investments in our Company, and senior management members of the Company	- Corporate governance - Risk and compliance management - Business ethics - Product and service safety and quality	- Financial and performance reports, Shanghai Stock Exchange e-interaction platform, emails, and roadshows - Risk management system - Anti-corruption mechanisms - Comprehensive audits - Quality reviews
Customers	IDMs and fabless semiconductor companies	- Information security and privacy protection - Product and service safety and quality - R&D innovation and intellectual property protection - Green products - Clean technology opportunities - Circular economy	- Information Security Management System (ISMS) - ISO9001 and other quality management certifications - Customer satisfaction surveys - Technical seminars and industry exchange meetings - Product lifecycle management
Employees	Our employees and contract personnel who serve our Company on a continuous basis	- Rights and Interests of Employees - Employee health and safety - Employee development and training	- Employee rights protection systems - Dual prevention mechanism for safety risk classification control and hidden danger investigation and management - Comprehensive training programs for all levels

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Key Stakeholders	Description	Concerned Topics	Engagement and Response Methods
Government and regulatory authorities	Tax, environmental protection, safety and other departments, local governments and regulatory authorities (such as SFC, etc.)	- Compliant operation - Emissions and waste management - Energy management - Water resources management - Climate change mitigation and adaptation	- Compliance and internal control systems - Carbon footprint reports - Water resource risk assessments in factory operations - Environmental information disclosure
Partners	Suppliers, research institutions, industry associations, etc.	- R&D innovation and intellectual property protection - Sustainable supply chain management - Circular economy	- Industry exchanges - Responsible Business Alliance (RBA) certification - Local supplier diversification - Supplier social responsibility audit mechanisms
Communities and the public	Communities in which we operate, the public and media, etc.	- Community and public welfare - Emissions and waste management	- Engagement in Community activities, employee volunteer activities, public welfare activities, social cause campaigns, etc. - Environmental information disclosure

The Company established and gradually improved due diligence mechanism, appointed the ESG Working Group to lead and coordinate various departments to carry out due diligence such as stakeholder communication, interviews and research, so as to analyze, identify and determine the impacts, risks and opportunities of ESG topics in the short term (within 1 year), medium term (1 year to 5 years) and long term (more than 5 years), so as to facilitate targeted response measures.

Hua Hong Semiconductor 2024 Topics Impacts, Risks and Opportunities Analysis

No.	Topics	Timeframe	Key Risks and Opportunities	Value Chain Scope		
				Upstream	Operations	Downstream
1	Climate Change Mitigation and Adaptation	Short, Medium, Long-term	Risk: Increasing compliance costs and low-carbon technology upgrade requirements due to stricter global climate and environmental policies and carbon emission control measures Opportunity: Investments in climate-friendly technologies and development of more efficient, eco-friendly products can enhance market competitiveness and brand reputation	√	√	√

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No.	Topics	Timeframe	Key Risks and Opportunities	Value Chain Scope		
				Upstream	Operations	Downstream
2	Energy Management	Short, Medium, Long-term	Risk: Optimizing energy management systems may incur additional costs and raise concerns from energy-efficiency-focused customers and investors Opportunity: Declining costs of renewable energy (e.g., solar, wind) provide stable, low-cost energy options, reducing reliance on fossil fuels	√	√	√
3	Water Resources Management	Medium, Long-term	Risk: Water shortages in production operations may lead to disruptions, unstable water quality, and reduced capacity and product yield, increasing costs and revenue losses Opportunity: Improved water resources management systems enhance efficiency and reduce operational costs		√	
4	Emissions and Waste Management	Medium, Long-term	Risk: Stricter emissions regulations may invest funds to ensure that its production processes and waste management meet new regulatory requirements, including upgrading treatment facilities or introducing new emission treatment technologies Opportunity: Higher waste recycling rates can transform waste into reusable resources, lowering raw material costs	√	√	
5	Environmental management	Medium, Long-term	Risk: Evolving environmental regulations require diligence and monitoring to ensure the operations comply with all relevant laws and standards. Non-compliance may result in fines, sanctions, or production restrictions Opportunity: Strong environmental practices can attract ESG-focused investors and customers, enhancing capital market image and market reputation	√	√	√

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No.	Topics	Timeframe	Key Risks and Opportunities	Value Chain Scope		
				Upstream	Operations	Downstream
6	Green Products	Short, Medium, Long-term	Risk: In a rapidly evolving technological environment, failure to update product designs and production processes may result in falling behind competitors and losing market share Opportunity: Effective product lifecycle management enables rapid response to market demands and technological advancements, driving product innovation and diversification		√	√
7	Circular Economy	Medium, Long-term	Risk: Without proper management and operational mechanisms for recycling raw materials and waste, product quality may be compromised Opportunity: Improved resource efficiency can reduce raw material procurement costs and waste disposal expenses, enhancing overall operational efficiency	√	√	√
8	Clean Technology Opportunities	Medium, Long-term	Risk: Rapid advancements in clean technology may render the products less competitive if they fail to keep pace Opportunity: Growing global demand for green and clean technologies and products presents opportunities to expand market share and create new revenue streams through innovation		√	√
9	Employee Health and Safety	Medium, Long-term	Risk: In high-intensity production environments, accidents and injuries may occur if safety measures are not effectively managed, potentially leading to employee injuries or even more severe accidents Opportunity: Strengthening occupational health and safety management can improve safety standards, reduce accidents, and mitigate legal and financial risks, creating a safer work environment	√	√	

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No.	Topics	Timeframe	Key Risks and Opportunities	Value Chain Scope		
				Upstream	Operations	Downstream
10	Rights and Interests of Employees	Medium, Long-term	Risk: The departure of key employees may result in the loss of critical industry knowledge and skills, impacting project progress and execution efficiency Opportunity: Standardized employment policies and a positive work environment can enhance talent retention, supporting sustainable operations and business growth		√	
11	Employee Development and Training	Medium, Long-term	Risk: Failure to provide high-quality training may lead to gaps in employees' professional knowledge and management capabilities, affecting product quality and managerial effectiveness Opportunity: High-quality training fosters innovative thinking and problem-solving skills, enabling the Company to adapt to market changes and technological advancements, thereby enhancing competitive advantage		√	
12	Product and Service Safety and Quality	Short, Medium, Long-term	Risk: Inadequate product quality and customer service may increase after-sales and maintenance costs, including additional labor costs and recall expenses, potentially leading to customer loss and revenue decline Opportunity: Robust product quality management systems and customer relationship management systems can enhance product competitiveness, improve customer satisfaction, and drive revenue growth through increased loyalty	√	√	√

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No.	Topics	Timeframe	Key Risks and Opportunities	Value Chain Scope		
				Upstream	Operations	Downstream
13	Information Security and Privacy Protection	Medium, Long-term	Risk: Strengthening cybersecurity infrastructure requires continuous financial and resource investments, including firewalls, intrusion detection systems, and data encryption technologies, to mitigate the risk of cyberattacks Opportunity: Effective information security management can attract privacy-conscious customers, enhance brand image, increase market share, and drive sales growth		√	√
14	Sustainable Supply Chain Management	Medium, Long-term	Risk: Ensuring supply chain stability and sustainability may require diversifying suppliers and incurring higher procurement costs to meet environmental and social responsibility standards, potentially impacting profitability and competitive advantage Opportunity: Building a sustainable supply chain enhances the Company's ability to identify and manage environmental and social risks, improving operational resilience and reducing potential business risks	√	√	
15	R&D Innovation and Intellectual Property Protection	Short, Medium, Long-term	Risk: Market acceptance of new technologies or products may fall short of expectations, affecting sales and reducing revenue and profitability Opportunity: Success in technological innovation can attract investor attention, and R&D achievements can create financing opportunities and open new business prospects	√	√	√

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No.	Topics	Timeframe	Key Risks and Opportunities	Value Chain Scope		
				Upstream	Operations	Downstream
16	Community and Public Welfare	Short, Medium, Long-term	Risk: If the Company's performance in public benefit activities or social responsibility fails to meet public expectations, it may harm brand image and reputation Opportunity: Involvement and investment in targeted communities can help the Company better understand local market needs, seize expansion opportunities, and strengthen competitive advantage		√	
17	Corporate Governance	Medium, Long-term	Risk: Non-compliant corporate governance may lead to regulatory investigations and fines, impacting financial health Opportunity: Strong corporate governance can attract more investors, enhance risk management capabilities, and foster trust, driving high-quality development	√	√	√
18	Business Ethics	Medium, Long-term	Risk: Violations of business ethics may result in legal disputes, hefty fines, and increased financial burdens Opportunity: Improving governance structures and transparency can strengthen internal controls and risk management, reducing the likelihood of violations	√	√	√
19	Risk and Compliance Management	Medium, Long-term	Risk: Failure to comply with relevant laws and regulations may lead to legal disputes, fines, or sanctions, damaging the Company's reputation and market trust Opportunity: Optimizing internal processes and systems can improve operational efficiency, reduce compliance risks, and support long-term business development while minimizing compliance costs	√	√	√

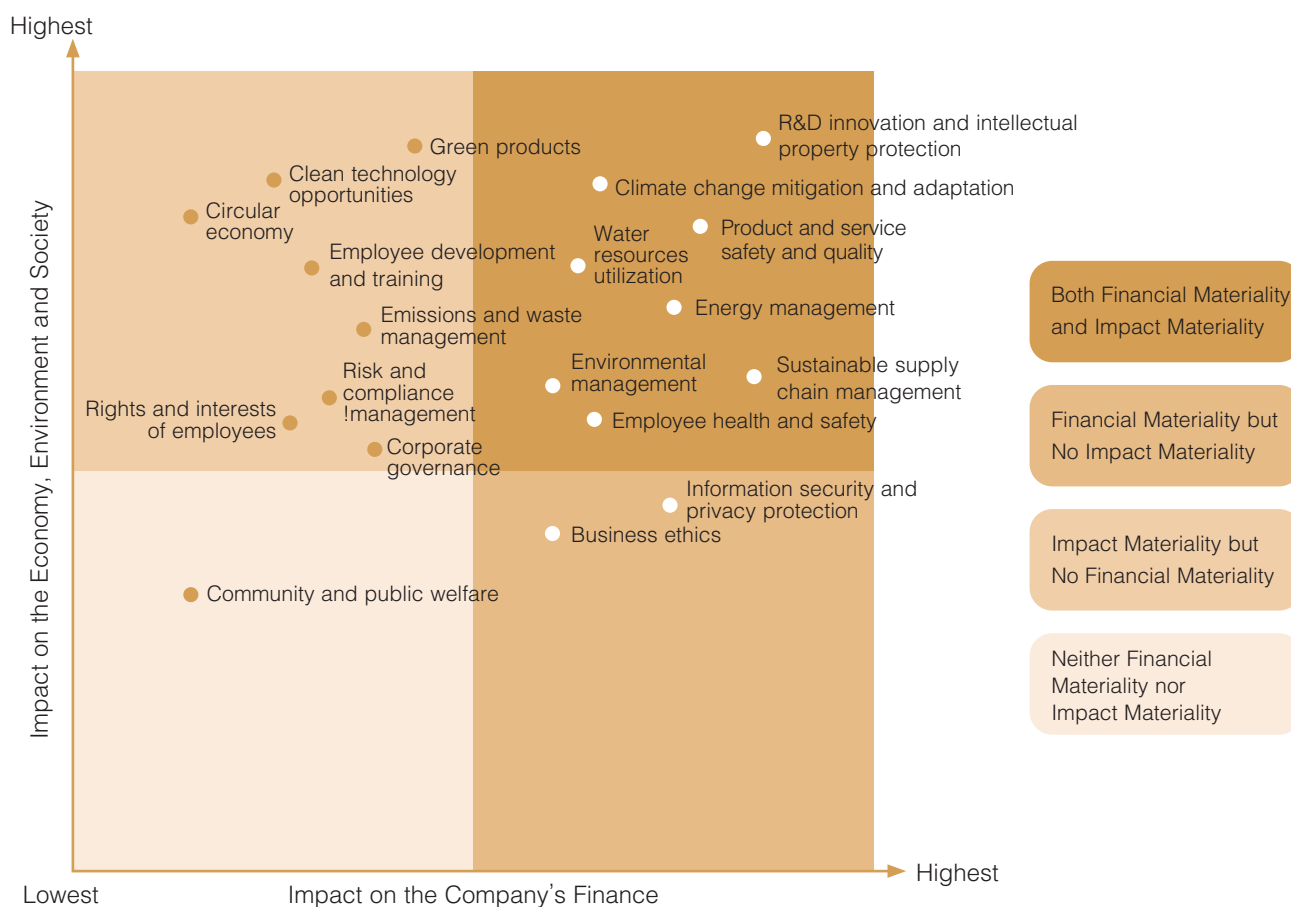
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1.3 Materiality Analysis Conclusion

In 2024, through the double materiality analysis, Hua Hong Semiconductor identified 19 material topics, categorized as follows: 7 topics with double materiality, 2 topics with financial materiality only, 9 topics with impact materiality only, and 1 topic with neither financial nor impact materiality. The specific categorization is as follows:

- Topics with double materiality: R&D innovation and intellectual property protection, climate change mitigation and adaptation, water resources management, energy management, environmental management, sustainable supply chain management, and employee health and safety;
- Topics with financial materiality only: Information security and privacy protection and business ethics;
- Topics with impact materiality only: Green products, clean technology opportunities, circular economy, employee development and training, product and service safety and quality, emissions and waste management, risk and compliance management, rights and interests of employees, and corporate governance;
- Topics with neither financial materiality nor impact materiality: Community and public welfare.

Hua Hong Semiconductor 2024 Topics Materiality Matrix



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Note: Among the 21 topics set in the SSE Guidelines, the Company does not involve the topics of “ecosystem and biodiversity protection” and “science and technology ethics”; “due diligence” and “stakeholder engagement” are processes for ESG topic identification, assessment and management, and are responded to in the “Materiality Assessment” section, not included in the matrix as ESG topics. The remaining Guideline topics are addressed in the main body of the report or in the index table. For the specific chapter mapping and explanation, please refer to the “Benchmarking Index Table of Self-Regulatory Guidelines No.14 for Companies Listed on Shanghai Stock Exchange—Sustainability Report (For Trial Implementation)”. In addition, the Company has independently identified the topics of “Green Products”, “Clean Technology Opportunities”, “Risk Compliance Management” and “Corporate Governance”, all of which are responded to in the main body of the report.

2. ESG Management System

2.1 ESG Management Structure

Based on the corporate vision of “Continuous Innovation and Empowering the Future for Global Customers”, the Company established a top-down environmental, social, and governance (ESG) management structure, continues to improve the ESG management system, encourages innovation-driven development, fosters innovative talents, and strives to reduce the impact of business operation on the environment. We also make efforts to diversify the supply chain capabilities and promote the joint creation of corporate business and social values.

The Board, as the highest decision-making/management body for the Company’s ESG management, is responsible for the following aspects:

- to direct the formulation of ESG management policies and strategies of the Company and ensure that they are up to date, relevant, and in compliance with applicable legal and regulatory requirements;
- to direct the identification and determination of the importance of significant ESG topics of the Company;
- to supervise the setting and implementation of the Company’s ESG goals, including: setting ESG management performance goals of the Company; monitoring progress in achieving the goals; and advising on actions required to achieve the goals;
- to review and approve the Company’s annual Environmental, Social, and Governance Report and other ESG related disclosures.

The management level is responsible for overseeing the overall implementation of the ESG objectives set forth by the Company. The ESG Working Group is tasked with identifying the impacts, risks, and opportunities associated with ESG topics, developing ESG strategies and management policies, and assisting relevant departments in implementing ESG management practices. It also regularly reports to the management team and the Board on the progress of key ESG performance indicators, thereby advancing the achievement of the Company’s ESG management objectives.

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ESG Management Structure



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2.2 ESG Management Strategies and Targets

The Company has integrated ESG management into its products, business operations, and corporate development and formed a complete ESG management strategy covering the four aspects of “Responsibility for Employees”, “Responsibility for the Industry”, “Responsibility for People’s Livelihood”, and “Responsibility for Investors”.

ESG Management Strategies



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The Company has set ESG management targets for water use efficiency, energy use efficiency, GHG emission reduction, pollutant reduction, waste reduction, and conflict mineral due diligence, based on the development of its business and its ESG performance in operations. On an annual basis, the management and the Board review the ESG performance and achievement of the Company's ESG goals for the previous year and disclose the results in the ESG report, so as to promote the achievement of ESG management targets.

Aspect	Target	Progress in 2024
Water resources management	The water consumed per unit product ¹ (m ³ per 8-inch wafer) in 2030 will decrease by 5% as compared with 2023	The water consumed per unit product was 2.76 m ³ per 8-inch wafer, decreasing by 2.49% as compared with 2023
Energy consumption management	Comprehensive energy consumption per unit product (MWh per 8-inch wafer) in 2030 will decrease by 10% as compared with 2023	Comprehensive energy consumption per unit product was 0.27 MWh per 8-inch wafer, decreasing by 9.08% as compared with 2023
GHG emissions management	GHG emissions per unit product (tCO ₂ e per 8-inch wafer) in 2030 will decrease by 10% as compared with 2023	GHG emissions per unit product in 2024 was 0.12 tCO ₂ e per 8-inch wafer, decreasing by 9.53% as compared with (0.14) in 2023
Waste discharge management	The waste produced per unit product (kg per 8-inch wafer) in 2030 will decrease by 4% as compared with 2023	The waste produced per unit product was 8.29 kg per 8-inch wafer, increasing by 8.41% as compared with 2023
Wastewater discharge management	COD discharge concentration in wastewater will be 30% lower than the emission standard	Shanghai Production Base: The average COD discharge concentration was 65.70 mg/L, which is 30% lower than the emission standard of 500 mg/L Wuxi Production Base: The average COD discharge concentration was 29.90 mg/L, which is 30% lower than the emission standard of 50 mg/L
Air emission discharge management	VOCs discharge concentration in air emissions will be 30% lower than the emission standard	Shanghai Production Base: The average VOCs discharge concentration was 5.35 mg/m ³ , which is 30% lower than the emission standard of 50 mg/m ³ Wuxi Production Base: The average VOCs discharge concentration was 0.33 mg/m ³ , which is 30% lower than the emission standard of 100 mg/m ³
Sustainable supply chain management	The due diligence rate on suppliers of conflict minerals will reach 100%, and the utilization rate of compliant minerals will reach 100%	Have conducted due diligence on suppliers of "conflict minerals", with a coverage rate of 100%, and completed the latest version of the investigation report on conflict minerals and extended minerals All suppliers are found to use compliant raw materials with the utilization rate of compliant minerals up to 100%

¹ Water consumed per unit product = water consumption from municipal water supply / annual production, excluding recycled water and other water sources.

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Honors and Recognitions Received by Hua Hong Semiconductor in 2024

No.	Honors
	Global
1	Responsible Business Alliance Platinum Award
	National
2	Second Prize in State Science and Technology Improvement Award
3	National Worker Pioneer
4	National Model Unit for Professional Ethics Construction
	Shanghai
5	Shanghai Quality and Technology Progress Award – Excellence Award
6	The 35th Shanghai Outstanding Invention Gold Award
7	Shanghai Green Manufacturing Demonstration List in 2024 (Second Batch) - Green Factory
8	Excellent Workstation of Academician in Shanghai
9	Shanghai Worker Pioneer
10	First Prize for Outstanding Collective in Shanghai Employee Labor and Skill Competition
11	Second Prize in Shanghai Employee Outstanding Innovation Achievement Award
12	Outstanding Achievement Award for Advanced Operational Methods of Shanghai Municipal Employees
13	Shanghai Employee Rationalization Proposal Innovation Award
	Jiangsu
14	Wuxi Model Unit for Happy Enterprise Construction
15	Wuxi City Model Worker's Home

3. Energy Conservation and Emission Reduction to Address Climate Change

3.1 Climate Change Mitigation and Adaptation

In the operational process of Hua Hong Semiconductor, the direct sources of GHG emissions are the burning of natural gas, gasoline, diesel and the fugitive emissions of the production process, and the indirect sources of GHG emissions are purchased heat and electricity. With reference to the International Sustainability Standards Board (ISSB)'s Sustainability Disclosure Standards of IFRS 2 – Climate-Related Disclosures (“IFRS S2”), the Company has established a top-down climate governance framework and formulated development strategies to cope with climate change, so as to enhance its adaptability to climate change and gradually minimize the impact of its operations on the environment.

Hua Hong Semiconductor's Climate Change Management System

Governance	- The Board regularly reviews climate-related risks and opportunities, as well as climate change response strategies and management targets, and regularly monitors progress towards the achievement of those targets. - The ESG Working Group takes responsibility for formulating climate change response strategies and management targets. According to the external regulatory requirements and market demands, the ESG Working Group makes timely adjustments to the timeliness of management policies, supervises the implementation of related initiatives at each plant, and regularly reports the work progress to the Board. - Each plant is responsible for executing specific tasks, such as verifying GHG emissions, reducing GHG emissions through practical actions, regularly collecting and summarizing GHG emission data, and performing self-assessments on the achievement of targets.
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Strategy

- Identifying climate change risks, including transition risks and physical risks, and conducting scenario analysis to evaluate the financial impacts of climate change risks.
 - a) Transition risks: Actively responding to a series of measures under the emission control policies to continuously reduce GHG emissions through various initiatives such as lean production, green procurement, improvement of production processes and replacement of non-essential energy-intensive equipment, details of which are available in the section headed “Energy Management”.
 - b) Market opportunities: Capitalizing on the growth opportunities of product application in wind and solar power generation, white goods and automotive electronics industries, details of which are available in the section headed “Clean Technology Opportunities”.
 - c) Product opportunities: Developing low-power green products and utilizing our process advantages to reduce the ESG emissions at the consumer end. Reducing the carbon footprint of products through whole life-cycle management, details of which are available in the section headed “Green Products”.
- Making clean technology innovation as one of our core strategies and incorporating climate change strategies into our business development frameworks, such as resource use efficiency, energy sources, products and service opportunities.

Impacts, Risks and Opportunities Management

- Integrating the management of climate-related risks and opportunities into the environmental risk management system, and formulating response strategies and implementing management measures for significant climate-related risks and opportunities.
- Conducting ISO14064-1 GHG emission verification and certification to systematically identify the GHG emission sources and verify the GHG emissions.

Metrics and Targets

- Publicly disclosing the GHG emission data annually through the ESG reports and reporting the management targets along with their achievement status.
- Management target of GHG emissions: GHG emissions per unit product (tCO₂e per 8-inch wafer) in 2030 will decrease by 10% as compared with 2023.
- Achievement status: GHG emissions per unit product in 2024 was 0.12 tCO₂e per 8-inch wafer, decreasing by 9.53% as compared with 2023.

The Company regularly identifies climate-related risks and opportunities, assesses their impacts on its business development and business model, and analyzes the potential financial impacts, so as to better adjust its strategies and management approach and continuously improve its response to climate change.

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Potential Financial Impacts on and Response Measures Taken by Hua Hong Semiconductor Towards Climate-related Risks and Opportunities

Risks and Opportunities of Climate-related as Identified		Potential Financial Impacts	Response Measures
Risks	Transition Risks Transitioning to a low-carbon economy, various stakeholders expect companies to take active actions in response to climate change and to enhance the transparency of information disclosure. As a result, any company that fails to respond effectively to these propositions from the stakeholders would be exposed to negative impacts on its own reputation. Reputation Risks The PRC has set a "Dual Carbon" target, and regulatory scrutiny on corporate carbon emissions will continue to intensify in the future. Companies whose carbon emissions do not meet the regulatory requirements shall purchase carbon emission allowances or China Certified Emission Reductions (CCERs), which will incur additional operating costs; companies that fail to achieve their carbon emission quota in full and on time may be subject to penalties such as rectification within prescribed period by relevant government departments. Policy and Legal Risks Customers and consumers are increasingly concerned about the sustainability of products and are demanding higher standards for carbon footprint and energy consumption. If the Company fails to meet the market demands for its products and services in a timely and effective manner, the Company will lose its market advantage. Market Risks	Decreased Revenue Increased Operating Costs Decreased Revenue	- Publicly disclosing the GHG emissions and reporting the target progress of GHG emissions per unit product annually. - Disclosing climate change mitigation and adaptation efforts in accordance with IFRS S2, and reporting the progress to stakeholders. - Conducting carbon accounting work annually, and completing the compliance and achievement of carbon emission quota limits required by relevant ministries and commissions in a timely manner. - The Company's annual carbon emissions do not exceed the carbon quota limits, therefore no additional costs are incurred. - Reducing GHG emissions through various measures, such as upgrading process technologies and enhancing production equipment performance. For example, use of NF_3 with high cleaning efficiency instead of C_2F_6 helps to reduce use of fluorocarbons, thus reducing GHG emissions annually. - Developing products with lower power consumption and higher efficiency to help downstream industries improve energy efficiency and reduce GHG emissions across the value chain.

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Risks and Opportunities of Climate-related as Identified		Potential Financial Impacts	Response Measures
Physical Risks	Acute Physical Risks The Company operates in coastal regions such as Shanghai and Wuxi and may be exposed to climate-related disasters such as typhoons and rainstorms, which may damage fixed assets, including infrastructure, resulting in economic losses.	Reduced Value of Fixed Assets	- Formulating emergency response plans for extreme weather events. - Regularly conducting emergency drills and training for natural disaster incidents.
	Chronic Physical Risks Persistent high-temperature weather caused by climate change may disrupt the Company's business operations, resulting in lower business revenue.	Decreased Revenue	
	Resource Use Efficiency Resource use efficiency shall be improved, including the efficiency of energy and water resources, which can help the Company reduce operating costs during operations.	Decreased Operating Costs	The 12-inch production line in Fab 7 (Wuxi) was awarded Leadership in Energy and Environmental Design ("LEED") Gold certification from the United States Green Building Council ("USGBC").
	Energy Sources A higher utilization rate of clean energy in production activities can help reduce reliance on fossil energy and, in the long term, reduce the cost of purchasing municipal electricity.	Decreased Operating Costs	Developing a rooftop photovoltaic construction plan to use renewable energy instead of non-renewable energy and increase the proportion of green electricity.
Opportunities	Products and Services The Company's products are widely used in energy solutions for household appliances and across all stages of the photovoltaic and energy storage industries, from electricity generation to consumption. The growing demand for chips in these sectors is driving a continuous increase in the Company's business revenue	Increased Revenue	The Company possesses non-volatile memory (eNVM), high-performance microcontroller (MCU) and single-chip integration (BCD) processes, and has accumulated rich technology and experience in the new energy power generation industry and the manufacturing processes in the field of home appliance applications.

During the Reporting Period, the Company completed the self-inspection report on the evaluation of dual-control responsibility for carbon emissions in accordance with the Notice on Conducting 2023 Pilot Evaluation of Dual-Control Responsibility for Carbon Emissions in Key Energy-Consuming Enterprises in the Industrial and Communication Industry 《關於開展 2023 年度工業和通信業重點用能單位碳排放雙控責任評價考核工作(試點)的通知》 issued by Shanghai Municipal Economic and Information Technology Commission. By implementing measures to reduce the GHG emissions from the Company's production and operation at both the management and equipment levels, the Company is progressively advancing toward its green and low-carbon transition objectives.

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The Company's Shanghai Production Base (Fab 1, Fab 2 and Fab 3) is a carbon emission quota management unit in Shanghai. The Shanghai Production Base conducts third-party verification of the GHG emission reports every year and completes the total data control plan for the next year at the end of each year. During the Reporting Period, the Company completed the compliance and settlement work on schedule, with a total payment quota of 251,024 tons.

3.2 Energy Management

Hua Hong Semiconductor is committed to building a green enterprise integrating green technology, green production and green building to realize environment-friendly operation and sustainable development. The Company's energy sources include electricity, heat, natural gas, gasoline and diesel, of which electricity, heat and natural gas are sourced from municipal supplies, with no significant impact on the surrounding environment and natural resources.

Governance

The Company has formulated various administrative systems, including the Resource and Energy Management Procedure and the Management Regulations for Water, Electricity, Steam, and Gas Metering in compliance with the Energy Law of the People's Republic of China and the Energy Conservation Law of the People's Republic of China. During the Reporting Period, the Company improved its energy management system in accordance with ISO 50001 and sorted out and updated its energy management system, to further standardize its energy management work.

The Company has established a three-tier energy management structure comprising the Board, the Management and the ESG Working Group. The Board is responsible for reviewing the energy management strategy and overseeing the achievement of targets. The Management is responsible for formulating the energy management strategy and targets, facilitating implementation of relevant work in each production base and department, overseeing achievement of targets, and reporting to the Board. The ESG Working Group leads the EHS, power and production departments in each production base to formulate specific action plans to implement and realize the management strategy.

Strategy

The Company positions energy conservation as an important strategic development direction. On the one hand, the Company has reduced total energy consumption through energy-saving technological transformation and actively complied with the energy consumption assessment requirements of the governmental or regulatory authorities, to reduce the policy risks. On the other hand, the Company has strategically deployed photovoltaic power generation facilities to safeguard against operational risks caused by interruption of the municipal energy supply, thereby avoiding additional operating costs resulting from energy shortages and price fluctuations.

The Company actively integrates the concepts of clean production and low-carbon energy into its production and operation processes and turns energy conservation achievements into product and service opportunities, to create green products with low environmental impacts. Through eco-design, residual heat recovery and energy-saving renovation, the Company is building green factories. In December 2024, Fab 1 was awarded the Shanghai Green Manufacturing Demonstration List in 2024 (Second Batch) - Green Factory.

Impacts, Risks and Opportunities Management

To address operational risks arising from energy supply interruptions, the Company has developed various operational protocols, such as the Emergency Plan for Emergency Unloading, the Emergency Plan for Emergency Power Cut, the On-site Treatment Plan of the Power Department for Power Cut and the Emergency Plan for Power Circuit Breaker Tripping. Emergency response training and emergency drills were carried out to improve the ability of its employees to manage various emergency power cut scenarios.

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In response to increasingly stringent local government regulations on energy consumption, the Company has formulated internal management targets that are higher than the regulatory requirements, to mitigate the impacts of policy risks. Aligned with the “Hundred One” Energy Conservation and Carbon Reduction Initiative in the Shanghai Industrial and Communication Industry (2022-2025), the Company rigorously implemented the “energy conservation project plan of 3,000 tons of standard coal”. By 2024, all 15 energy-saving renovation projects identified in the 14th Five-Year Plan energy audit work were fully completed, achieving an energy reduction of 3,339 tons of standard coal.

Metrics and Targets

The Company has set an overall energy management target, which is divided up and allocated to each production base for implementation. The Company monitors energy consumption of each production process on a daily basis and regularly reviews the progress towards achieving the target.

Hua Hong Semiconductor’s Energy Management Metrics and Targets in 2024

Metric	Unit	2023 Performance	2024 Performance	2030 Target	Progress Towards the Completion of the 2024 Target
Comprehensive energy consumption per unit product	MWh per 8-inch wafer	0.30	0.28	Decreasing by 10% as compared with 2023	Decreasing by 5.91% as compared with 2023 – in progress of completing the target
Honor and Award	• Awarded Shanghai Green Manufacturing Demonstration List in 2024 – Green Factory				

In order to effectively enhance energy efficiency management of equipment, the Company conducted energy consumption statistics on major energy-consuming equipment and formulated a work plan for energy efficiency upgrading and equipment renovation, in accordance with the Notice on Conducting Energy Efficiency Benchmarking and Renovation on Major Energy-Consuming Equipment of Enterprises in the Industrial and Communication Industry (《關於開展工業和通信業企業重點用能設備能效對標及更新改造的通知》) issued by the Shanghai Municipal Economic and Information Technology Commission.

Hua Hong Semiconductor’s Energy Saving Initiatives and Achievements in 2024

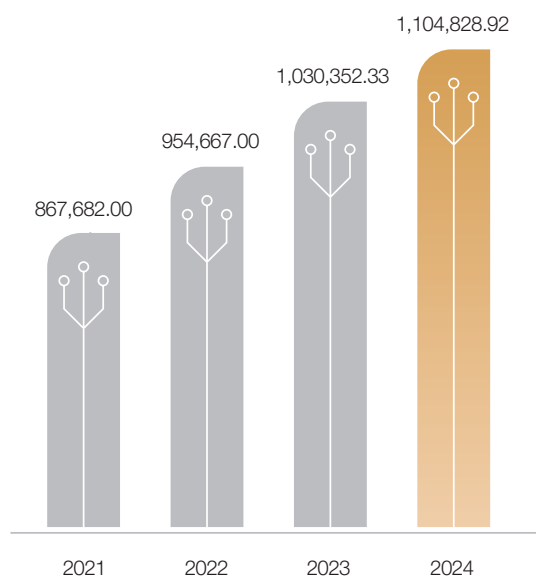
Entity	Specific Measures	Achievements
Shanghai Production Base	Variable frequency renovation on the RO high-pressure pumps in the pure water system	• Annual energy reduction of 28.23 tons of standard coal
	Reducing process steps by 3% and manufacturing cycle by 4% in the eFlash platform	• Annual electricity saving of 960,000 kWh
	Building an intelligent energy management platform Creating high-efficiency cooling rooms	• Annual electricity saving of 3.2 million kWh
	Completing heat recovery renovation on chilled water in the refrigeration unit	• Annual natural gas reduction of 188,000 m ³

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During the Reporting Period, the Company's primary energy consumption was as follows:

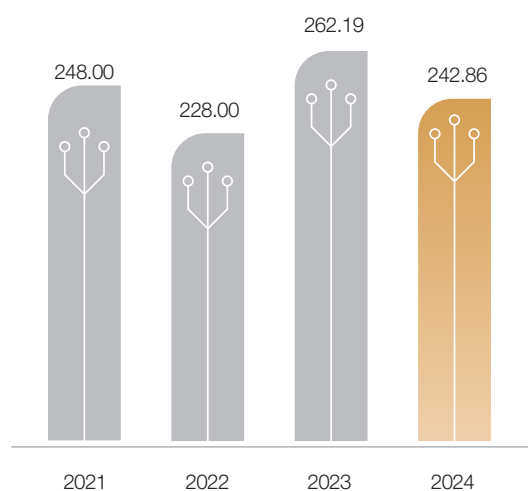
Total electricity consumption

Unit: MWh



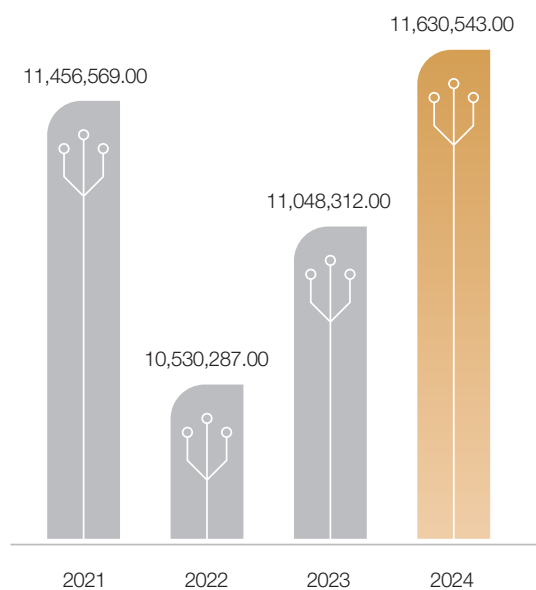
Electricity consumption intensity

Unit: kWh per 8-inch wafer



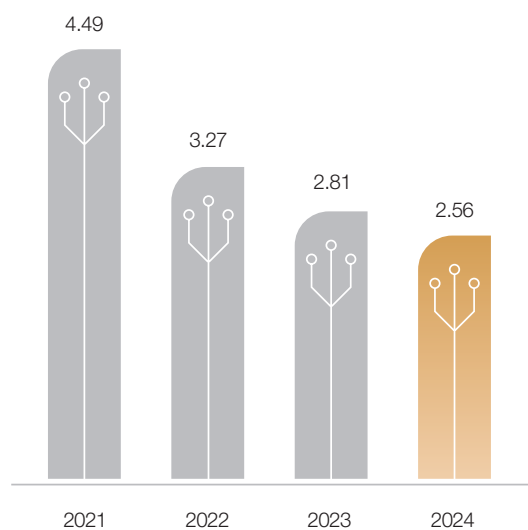
Total natural gas consumption

Unit: m³



Natural gas consumption intensity

Unit: m³ per 8-inch wafer



Note: The production expansion project of the Wuxi Production Base was put into operation in 2023, and the production capacity was expanded in 2024. The total consumption of electricity and natural gas has increased, however, the energy consumption intensity indicators have all decreased in 2024.

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3.3 Clean Technology Opportunities

Hua Hong Semiconductor actively responds to the “dual-carbon” economy call and solidly promotes its “8-inch + 12-inch” strategy and its advanced “Specialty IC + Power Discrete” dual-pronged strategy, to fully support the ecosystem construction across the whole industrial supply chain. Through continuous R&D and innovation, and expansion of specialty process advantages, Hua Hong Semiconductor focuses on the intelligent terminal, communication, Internet of Things, automotive electronics and other fields. Its business not only covers high-end consumer products, from IPM modules to peripheral control and power management, but also covers the power industry supply chain, from power generation to power consumption, as well as the five major chip systems in the field of automotive electronics.

The Company focuses on clean technology innovation as one of its core strategies and continues to promote development of new ecosystems and novel applications, such as embedded non-volatile memory, power management technology, power discrete technology and flash memory technology. This helps its customers to further expand their strategic positioning in new energy sectors, including automotive electronics, premium home appliances, photovoltaic systems and wind energy storage, and capitalize on market opportunities driven by clean technologies, thus actively contributing to future development of their business.

Products Supporting Clean Technology Development

Power Discrete

Power components is an important equipment component for new energy generation and new energy applications, such as photovoltaic and wind power generation equipment, electric vehicles, electric two-wheeled vehicles, the need to use a large number of insulated gate bipolar transistor (IGBT), metal – oxide semiconductor field effect transistor (MOSFET) and other devices.

The Company has accumulated a large number of customers in this field, and maintained a good relationship and deep co-development cooperation with the head enterprises in the field for a long time. So far, the scale of power discrete revenue has become the largest part of the Company's share.

Non-Volatile Memory

Non-volatile memory is widely used in chip applications such as microprocessors (MCUs) and application-specific integrated circuits (ASICs), while these types of chips are widely used in new energy generation equipment.

According to IC Insights, the automotive market accounts for more than 30% of the global MCU consumption market. All Hua Hong Semiconductor manufacturing plants have passed IATF 16949 automotive industry quality system certification, making important contributions to supporting local manufacturing of automotive semiconductors.

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New energy components and application fields involved

Product	Application Field	Percentage of Revenue
Power discrete	Photovoltaic and wind power generation equipment, electric vehicles, electric two-wheeled vehicles, etc.	31.0%
Embedded Non-Volatile Memory	New energy power generation equipment, electric vehicles, etc.	26.3%

Assist the development progress of new energy transformation technology

Automotive Electronics	Industrial New Energy	High-end Consumer Electronics
- Through supply chain collaboration, it helps customers accelerate their products into relevant module factories and complete machine enterprises. - Establish direct cooperative relationship with domestic mainstream automobile enterprises, from mature chip replacement to joint efforts to tackle key and difficult projects, and jointly build sustainable development ecosystem and other multi-dimensional cooperation. Recommended mature chip types cover various fields such as automotive powertrain, body control, cockpit, chassis and safety, remote control and communication, ADAS, charging piles, etc., to achieve mass supply. - A company-level automotive electronic database has been established and continuously improved.	The key products of power discrete have achieved mass supply and significantly increased market share. In the future, with the layout and strong support of the state for the double-carbon economy, the future growth of the new energy market can be expected.	- Cooperate with domestic well-known household appliance brand companies to improve domestic production replacement rate and continuously promote the expansion of high-end consumer electronics business. - Famous domestic household appliance brand companies have adopted IGBTs, display touch control chips, fingerprint lock chips and motor control products produced by the Company's technology platform, and have gradually penetrated into the air-conditioner, refrigerator and other major appliance markets.

The Company will further promote the development of silicon-based power discrete capable of achieving higher energy efficiency, and initiate the development of compound semiconductor power discrete with high power density and low energy consumption, so as to provide continuously optimized solutions for products in clean energy application fields such as inverter appliances and new energy vehicles, and further improve energy efficiency.

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4. Lean Production, Pioneering a Green Future

4.1 Environmental Management

Hua Hong Semiconductor's production and manufacturing fabs include the Shanghai Production Base and the Wuxi Production Base. Among them, Fab 1, Fab 2, Fab 3, and Fab 7 are key units monitored for water environment and environmental risks, while Fab 3 and Fab 7 are key units monitored for atmospheric environment. The Company discloses environmental information annually, in compliance with legal requirements.

The Company strictly adheres to the Environmental Protection Law of the People's Republic of China and the relevant environmental regulations of local governments and has established internal policies, such as the Standard Operating Procedures for Industrial Water Supply System, Internal Control Standards for Pollutants, and Operating Norms for Industrial Waste Management.

Governance

The ESG Working Group is responsible for formulating environmental management strategies and objectives, promptly tracking regulatory requirements from local governments and industry standards, identifying environmental management risks within the Company, overseeing environmental compliance in the daily operations of each fab, and reporting on key matters to the management team.

The EHS Department is responsible for coordinating establishment of the Company's environmental management system, including formulation of management regulations, performance analysis, and evaluation, with respect to energy and resource use, emissions management, etc. It also assists external agencies in the audit and testing of the Company's environmental management system.

Strategies

In terms of wastewater and air emission management, the Company prioritizes continuous improvement in environmental governance, with up-to-standard discharge as a key operational objective, to avoid operational risks caused by non-compliant discharge. The Company is located near office buildings and residential areas, where a significant number of people gather, so the Company ensures the normal operation of exhaust gas treatment facilities, avoids the spread of harmful substances to the surrounding environment, and creates potential exposure risks to the respiratory system and ecosystem of community residents.

The Company identifies risks throughout the lifecycle of pollutants, from generation and collection to treatment and discharge. Through regular monitoring and inspections, it ensures the proper operation of environmental protection facilities and compliance with air emission and wastewater treatment standards. Annually, the Company allocates a budget for environmental management and implements environmental protection and technical transformation projects, such as upgrading photoresist equipment and wastewater recycling systems. These initiatives aim to reduce raw material usage, extend the lifespan of chemicals, and minimize wastewater and air emissions, thereby mitigating the environmental impact of operations and production.

The Company emphasizes the capability of its staff to handle environmental emergencies, aiming to prevent environmental risk incidents, such as chemical leaks, fires, or direct pollutant discharges due to mismanagement. The Company regularly updates the emergency response plans and alternatives for environmental incidents at all fabs, formulates an annual emergency drill schedule each year and conducts emergency drills for potential environmental incidents. Fab 1 conducted on-site response drills for unexpected shutdowns of exhaust treatment systems, Fab 2 conducted on-site response drills for waste liquid leaks, Fab 3 conducted drills for SiF₄ cylinder leaks, and Fab 7 conducted drills for chemical spills.

Additionally, the Company has established a green office strategy to strengthen employees' environmental awareness through cultural initiatives, daily behaviour guidelines, new employee orientation and advocacy. This approach promotes the implementation of green office practices.

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Hua Hong Semiconductor's Practice in Green Culture

Saving paper around the office	- Implementing office informatization to reduce paper consumption. - Making efforts to promote the use of recycled paper, including for printing business cards of employees, to support recycling.
Saving electricity	- Regularly holding environmental management sharing sessions to strengthen employees' awareness of economical use of office resources and energy. - Turning off the power supplies when leaving work or a long absence. - Reasonably setting air conditioning temperature above 26°C in summer and below 20°C in winter.
Saving water	Developing employees' awareness of saving water and posting water saving signs.
Waste disposal	- Classifying domestic waste and implementing waste classification and disposal procedures, in accordance with the requirements at the place of operation. - Actively promoting waste classification initiatives to enhance employees' environmental awareness.
Commuting	- Encouraging employees to commute by public transport. - Using new energy vehicles as shuttle buses for the Company.

Impacts, Risks, and Opportunities Management

The pollutant discharge standards in the region and industry the Company operates in are continuously updated. To proactively mitigate policy risks, the Company has adopted internal pollutant discharge standards that are stricter than the regulatory requirements. In 2024, the local government of Shanghai updated the Discharge Standards of Pollutants for the Semiconductor Industry. In response, the Company's four fabs each established Internal Control Standards for Pollutants, to prepare for increasingly stringent environmental regulatory requirements.

The Company regularly identifies and assesses environmental risk topics at its production sites and refines its environmental management system. All four fabs have obtained ISO14001: 2015 certification. During the Reporting Period, no violation against environmental laws and regulations occurred. During the Reporting Period, the Company fully paid relevant taxes with respect to environmental protection.

Environmental Management System Certification Status of Hua Hong Semiconductor

Entity	Certification Name	Validity of Certification
Shanghai Production Base	ISO 14001: 2015 Environmental Management System	18 April 2027
Wuxi Production Base	ISO 14001: 2015 Environmental Management System	18 April 2027

The Company uses its safety management auditing and tracking system to analyze, sort out, and list the nonconforming items against environmental protection regulations found in daily management, with follow up and record keeping of subsequent rectifications through the regular reminder function of the system.

Metrics and Targets

The Company has set overall targets for environmental management, with the EHS department leading regular monitoring and evaluation to assess progress toward these targets and the actual effectiveness of control measures. In 2024, the Company invested RMB112.56 million in safety, environmental protection, and technical transformation projects, as part of our continuing commitment to improving environmental management facilities.

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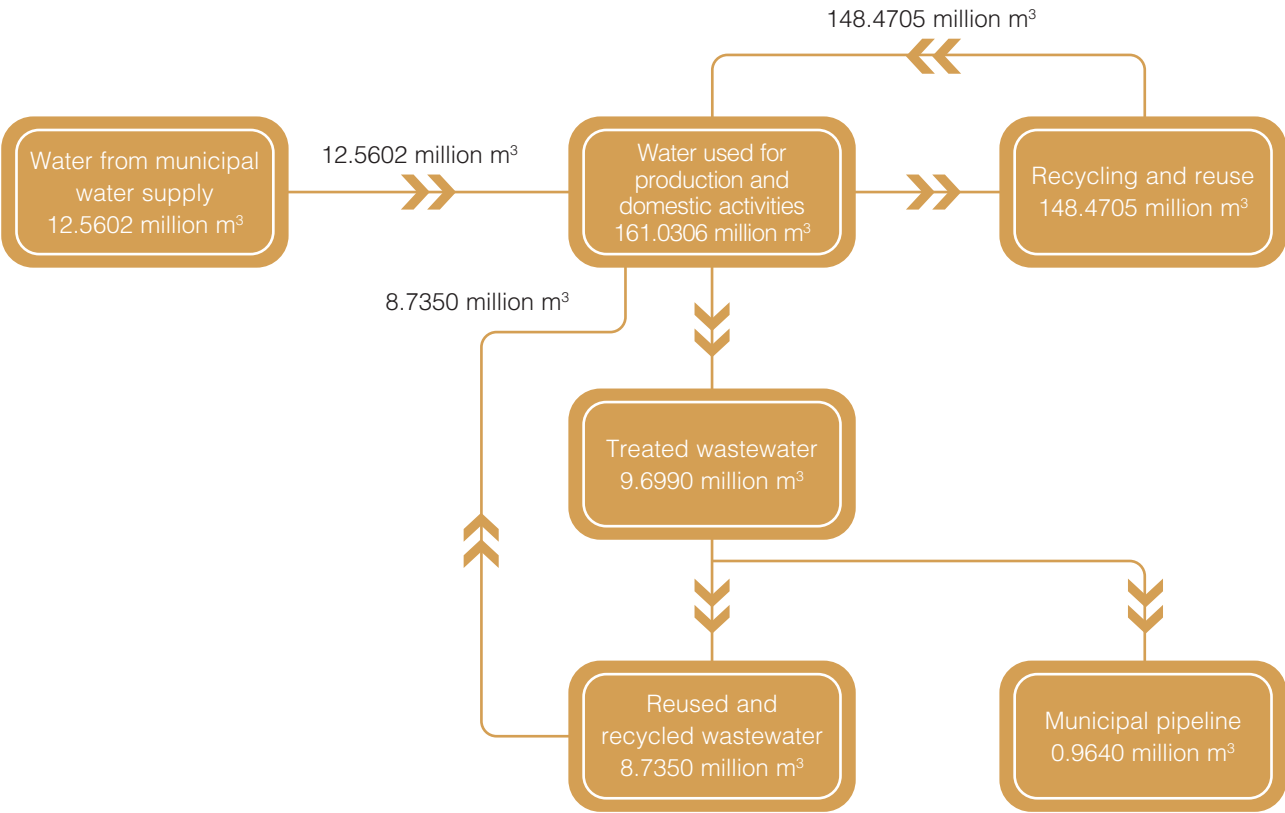
2024 Environmental Management Metrics and Targets of Hua Hong Semiconductor

Metric	Unit	2024 Performance	2024 Target
Number of environmental violations	Cases	0	0
Percentage of production base certified with ISO 14001	%	100	100
Total environmental expenditure	RMB	95,057,129	–

4.2 Water Resources Management

Water used by the Company comes to a large extent from the municipal water supply, in addition to a smaller amount of recycled water and air conditioning condensate from the pure water manufacturing process. In strict accordance with laws, rules and other relevant regulations, including the Water Law of the People’s Republic of China, the Shanghai Administrative Measures for Water Conservation, and the Jiangsu Water Conservation Ordinance, the Company has established management policies, including the Standard Operating Procedures for Industrial Water Supply System and the Tap Water Anomaly Handling Procedure, to monitor the supply and quality of water resources in its operational areas. The Company implements water-saving technical transformation, recycles production wastewater, and organizes water conservation awareness activities, to reduce water waste both in mindset and practice.

Overview of Water Used for Production and Operation of Hua Hong Semiconductor in 2024



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Governance

The Company has implemented a three-tier water conservation management system, comprising the “Fab Manager – Water Conservation Leadership Group – Water Conservation Management Team.” Led by the fab manager, the water conservation leadership group establishes water resource management objectives for the fabs, develops water usage regulations, and ensures that all departments adopt effective measures to achieve these objectives and comply with the policies. Additionally, each fab has established a water conservation management team, which designs water-saving initiatives tailored to production processes and operational workflows, oversees daily operations, monitors water consumption, and collects and reports data on water consumption.

Strategy

Semiconductor manufacturing requires water purity of high standards and consumes substantial amounts of water resources during production. As a water-intensive enterprise, the Company utilizes water across various production stages, some of which involving the use of pure water.

Recognizing the critical importance of a stable supply of water and pure water for on-going production, the Company has enhanced water use efficiency and increased investments and initiatives in water recycling. These measures aim to mitigate potential challenges such as water outage and unstable water quality arising from increased water costs and water scarcity, thereby ensuring stable production and maintaining high product yield.

To mitigate operational risks related to water resources, the Company regularly identifies and monitors water-related risks in its operations, evaluates the likelihood and impact of physical and policy risks, and develops corresponding management policies and response strategies.

Impacts, Risks and Opportunities Management

Through energy conservation and emission reduction initiatives, the Company continuously enhances the efficiency of water recycling, reduces its dependence on externally sourced ultrapure water and municipal water, and generates ultrapure water using RO reverse osmosis membranes. Furthermore, the Company actively explores water-saving opportunities and recycles water resources across various production stages.

The Company constantly monitors the risk of water resources in areas where we operate through the “Aqueduct Water Risk Atlas”, an external tool from the World Resources Institute (WRI). We analyze the rationality of water consumption for production and the operation of our fabs along with the possible impact of the water intake and implements measures to mitigate water-related risks.

WRI Risk Monitoring Results and Countermeasures of Hua Hong Semiconductor

Water risk monitoring indicators	• Water quantity risks: Water stress, groundwater level decline, interannual variability, seasonal variability, drought, river flood risks. • Water quality risks: Coastal eutrophication risk, exposure to untreated wastewater. • Legal, regulatory and reputational risks: Regulatory and reputational risks, lack of access to drinking water, sanitation risks, ESG reputational risks.
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Monitoring results of Shanghai Fab and Wuxi Fab	Water quantity risks	• Overall water quantity risk: Very high • Water stress: Very high • River flood risk: High • Drought risk: Medium to high • Groundwater level decline risk, annual and seasonal variability: Low to medium • Coastal flood risk: Low
	Water quality risks	• Overall water quality risk: Medium to high • Exposure to untreated wastewater: High • Sanitation risks: High • Coastal eutrophication risk: Low to medium
	Legal, regulatory and reputational risks	• Overall legal, regulatory and reputational risks: Medium to high • Regulatory and reputational risks: Medium to high • Lack of access to drinking water: Low to medium • ESG reputational risks: Low to medium
Countermeasures		• Relying on water monitoring sites for each fab, the Company has conducted water balance tests on a regular basis to calculate the process water, recycled water, wastewater, and domestic water use of each fab. • The Company records the flow, volume, and recycling of water resources, so as to analyze whether there is any abnormal utilization of water resources and predict the reasonable allocation of water resources among water-consuming units. • Upgrade or replace equipment with high water consumption, and enhance water consumption efficiency. • Expand the sources of water resources by reusing up-to-standard treated production wastewater and recycling by-products, such as concentrated water from pure water production and air conditioning condensate.

Metrics and Targets

The Company has set overall water resource management targets, which are allocated to each production base for execution. Water usage across all production processes is monitored daily, and progress toward achieving the targets is reviewed periodically.

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2024 Water Resource Management Metrics and Targets of Hua Hong Semiconductor

Metric	Unit	2023 Performance	2024 Performance	2030 Target	Progress towards the Completion of the 2024 Target
Water consumption per unit product	m ³ per 8-inch wafer	2.83	2.76	Decreasing by 5% as compared with 2023	Decreasing by 2.49% as compared with 2023 – In progress of completing the target
Honor and Award	Awarded the title of Advanced Energy-Conservation and Emission-Reduction Enterprise in Xinwu District, Wuxi.				

The Company improves water use efficiency through three measures, including water conservation, recycling production wastewater for processes with lower water quality requirements, and reusing treated production wastewater. Water-saving measures have been implemented in critical water usage areas and integrated wastewater recycling and treatment systems have been established to classify, filter, and reuse production wastewater.

2024 Water-Saving Measures and Achievements of Hua Hong Semiconductor

Entity	Action	Measures	Annual Water-Saving Results
Shanghai Production Base	Wastewater reuse	Concentrate and recycle reverse osmosis concentrate; treat and reuse equipment rinse water, instrument water, and condensate.	A total of 250,000 tons of water was saved during the year
Wuxi Production Base	Wastewater reuse	Concentrate and recycle reverse osmosis concentrate; treat and reuse equipment rinse water, instrument water, and condensate.	A total of 420,000 tons of water was saved during the year
	Wastewater recycling system upgrade	Upgrade backwash wastewater recycling system to increase the usage of recycled water.	A total of 2,500 tons of water was saved during the year

4.3 Emissions and Waste Management

Air Emission Management

Air pollutants generated during the production process of the Company include sulfuric acid mist (H₂SO₄), hydrogen chloride (HCl), nitric oxide (NO_x), ammonia (NH₃), and volatile organic compounds (VOCs).

Hua Hong Semiconductor strictly adheres to the Law of the People's Republic of China on the Prevention and Control of Atmospheric Pollution, the Discharge Standards of Pollutants for the Semiconductor Industry set by the relevant industry association of Shanghai and other applicable laws, regulations, and requirements. During the Reporting Period, each of the four fabs developed its own Internal Control Standards for Pollutants, adopting stricter internal discharge control measures that exceed regulatory requirements.

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Types and Treatment Methods of Air Pollutant of Hua Hong Semiconductor

Type	Treatment Method	
Air emission	Acid air emission	Up-to-standard discharge after removing most of the components through pretreatment and centralized treatment of the washing tower.
	Alkaline air emission (mainly ammonia gas)	Purification through a washing tower.
	Organic air emission	Purification through activated carbon adsorption, or combustion after concentration.
	Air Emissions containing dust (mainly small particulates of silicon dioxide)	Removal through a dust-extraction unit.

The Company formulates an annual testing plan, deploys online monitoring facilities, and performs regular maintenance of air emission treatment systems, to ensure effective treatment efficiency and compliance with air emission discharge standards. The Company proactively addresses regulatory requirements by gradually promoting the use of low-VOCs cleaning agents, thereby reducing untreated air emissions of volatile organic compounds during cleaning processes. In 2024, the Company's air emission testing results confirmed that all air emissions met discharge standards.

2024 Air Emission Management Metrics and Targets of Hua Hong Semiconductor

Metric	Unit	Production Base	2024 Performance		Progress towards the Completion of the 2024 Target
				2024 Target	
Annual average VOCs discharge concentration	mg/m ³	Shanghai Production Base	5.35	Better than the emission standard (50) by more than 30%, that is, <35	Emission concentration is more than 30% better than the emission standard, achieving target
		Wuxi Production Base	0.33	Better than the emission standard (100) by more than 30%, that is, <70	

2024 Air Pollutant Reduction Measures and Achievements of Hua Hong Semiconductor

Entity	Specific Measures	Results
Shanghai Production Base	Extended the chemical lifespan of SPM tanks (H ₂ SO ₄ +H ₂ O ₂) through research	Reduced annual sulfuric acid usage by 57 tons
Wuxi Production Base	Reduced SPM usage at cleaning stations	Reduced annual sulfuric acid mist emissions by 0.66 tons
	Upgraded boilers with low-carbon burners	Reduced annual nitric oxide emissions by 0.94 tons

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Wastewater Discharge Management

The Company strictly complies with the Law of the People's Republic of China on the Prevention and Control of Water Pollution and the Integrated Wastewater Discharge Standard. The Shanghai Production Base complies with the Discharge Standard of Water Pollutants for the Electronic Industry, while the Wuxi Production Base follows the Jiangsu Discharge Standards for the Semiconductor Industry. The Company has formulated internal management systems, such as the Operation Manual for Wastewater and Drainage Discharge and the Internal Control Standards for Pollutants. Through enhancements in wastewater treatment processes and the expansion of wastewater reuse initiatives, the Company has minimized the generation of wastewater pollutants.

Both the Shanghai Production Base and the Wuxi Production Base are equipped with online monitoring systems for wastewater pollutants, with discharge concentrations transmitted in real-time to the environmental regulatory authorities' public information platform, ensuring reliable, transparent, and traceable environmental information.

Types and Treatment Methods of Water Pollutants at Hua Hong Semiconductor

Type	Treatment Method	
Wastewater	pH, COD, ammonia nitrogen, fluoride ion, etc.	Discharged into the municipal sewage network upon meeting the discharge standards after being treated through a wastewater treatment system utilizing processes such as coagulation sedimentation, air stripping, acid washing absorption, aerobic biological contact, acid-base neutralization, MBR, etc., and then further treated by urban sewage treatment plants before being released into natural water bodies

During the Reporting Period, the wastewater test results of the Company indicated that all types of wastewater pollutants achieved compliant discharge.

Water Pollutant Management Metrics and Targets of Hua Hong Semiconductor in 2024

Metric	Unit	Base	2024 Performance	2024 Target	2024 Progress Towards Target
Annual COD discharge concentration	mg/L	Shanghai Production Base	65.7	Better than the emission standard (500) by more than 30%, that is, <350	Emission concentration is more than 30% better than the emission standard, achieving the target
		Wuxi Production Base	29.9	Better than the emission standard (50) by more than 30%, that is, <35	

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Measures and Achievements in Reducing Water Pollutant Discharges at Hua Hong Semiconductor in 2024

Specific Measures	Achievements
Recycling and reusing relatively clean cleaning wastewater to reduce wastewater discharge from hydrofluoric acid and deionized water (DHF) solution cleaning	Reducing DHF wastewater discharge by 10,000 tons/year
Completing the switch to 55LP Plus eFlash through the 55LP eFlash platform, reduce process steps by 3% and shorten manufacturing cycles by 4%	Reducing wastewater discharge by 3,708 tons/year

Waste Management

The Company complies with the Law of the People's Republic of China on the Prevention and Control of Environmental Pollution by Solid Wastes and the Standards for Pollutant Control in the Storage of Hazardous Wastes and has formulated management systems, such as the Waste Management Procedure and the Code of Practice for Industrial Waste Management, to standardize the processes of waste generation, temporary storage, transfer, and disposal. The Company also registers the waste disposal methods and destinations through the unified environmental protection management system of the government to ensure traceability. The waste is properly recycled, landfilled or incinerated by qualified transportation and disposal units.

During the Reporting Period, in accordance with the Notice on Further Strengthening the Standardized Environmental Management of Hazardous Wastes in the City issued by Shanghai Municipal Bureau of Ecology and Environment, the Company established an electronic tag application management system for hazardous wastes, achieved real-time connection with the solid waste system of Shanghai, and added the Management Measures for the Electronic Tag Application System for Hazardous Wastes, to standardize the electronic management of hazardous waste tags.

Types and Treatment Methods of Wastes at Hua Hong Semiconductor

Type		Treatment Method
Non-hazardous waste	Domestic waste and kitchen garbage	Entrusting the environmental sanitation departments in the places where we operate for unified disposal on a regular basis
	Raw materials for production and office supplies	Recycling within plant premises
	Sludge, etc.	Landfilling, and brick manufacturing
Hazardous waste	Waste acid, waste isopropanol, waste phosphoric acid, and organic waste liquid, etc.	Handled by suitable enterprises, or reused in other industries, for a higher reuse rate
	Waste glass bottles, 200L chemical barrels, and waste liquid from laboratory, etc.	Chemical treatment
	Cleaning cloths, plastic bottles, waste activated carbon, waste resin, and arsenic-containing waste, etc.	Incineration

The Company identifies and categorizes wastes, as well as adopts appropriate management measures and reuse methods, to strive in maximizing the recycling rate of wastes.

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Waste Management Metrics and Targets of Hua Hong Semiconductor in 2024

Metric	Unit	2023 Performance	2024 Performance	2030 Target	Progress towards the Completion of the 2024 Target
Waste generation per unit product	kg/8-inch wafer count	7.65	8.29	Decrease by 4% compared to 2023	Increase by 8.41% compared to 2023 towards the target

4.4 Circular Economy

The main types of raw materials for the Company's production and operation include silicon slices, quartz, target materials, chemicals, etc. To standardize the use of raw materials, the Company has formulated raw material management systems, such as the Raw Materials Shelf Life Control Policy, the Risk Identification and Response Measures for Key Materials, and the Key Materials Risk Analysis Table, ensuring that raw materials are within their shelf lives and that there are no quality or safety issues with key materials. For production auxiliaries, such as chemicals and gases, the Company has established the Materials Management Department's Management Standards for Chemical Storage and Gas-based Operations, promoting safe storage and handling procedures.

On the basis of standardizing raw material management, the Company, adhering to the principles of "reduction from the source" and "circular economy" and complying with the Law of the People's Republic of China on the Promotion of Circular Economy, strives to minimize the use of raw materials, especially chemicals, to further reduce safety risks. Additionally, it reduces waste of resources by improving production processes and rationally allocating resources for recycling.

Measures and Achievements in Saving Raw Materials at Hua Hong Semiconductor in 2024

Entity	Specific Measures	Achievements
Shanghai Production Base	Implementing modifications to the equipment used for photoresists, to reduce raw material consumption	Annual photoresist reduction by about 1,000 liters
	Improving production processes and extending the service life of chemicals	Annual savings of 9,100 liters of ammonia water and 4,300 liters of hydrogen peroxide
	Carrying out thin film process renovation	Annual savings of 1,217 kilograms of hexafluoroethane
Wuxi Production Base	Adjusting process parameters and reducing cleaning frequency	Annual savings of 2,520 liters of sulfuric acid Annual savings of 420 liters of hydrogen peroxide

4.5 Green Products

Chemical Control

Multiple processes in the Company's manufacturing operations require the use of chemicals, among which some hazardous and toxic chemicals require special management.

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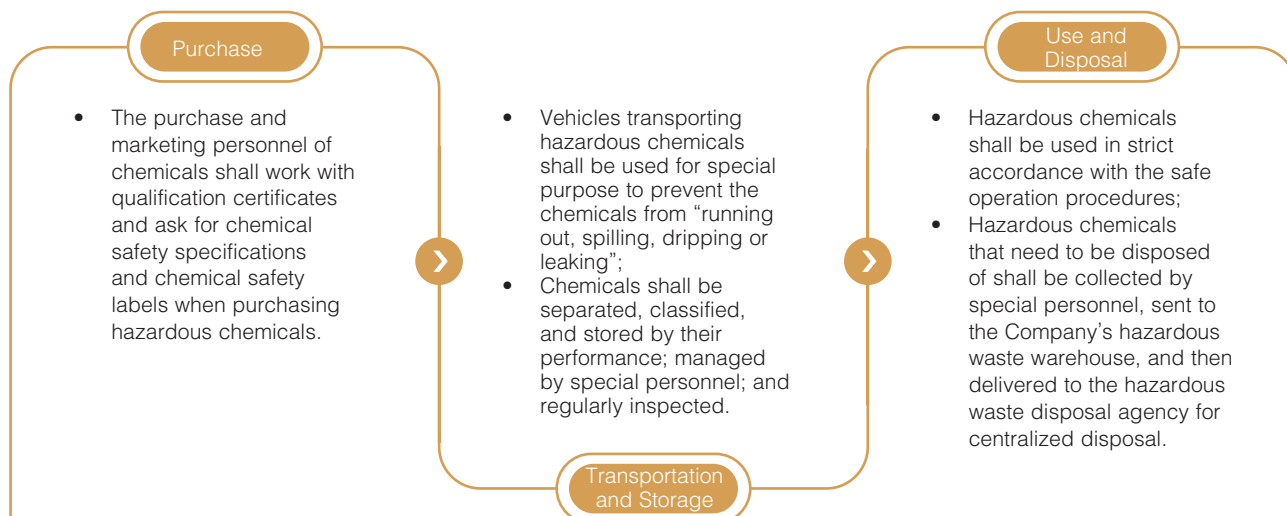
Types of Hazardous Chemicals of Hua Hong Semiconductor

Type	Name
Flammable liquid	Isopropanol, photoresist, and diesel fuel
Oxidants and organic peroxides	Hydrogen peroxide
Toxic chemicals	Phosphine and fluorine
Corrosive chemicals	Sulfuric acid, hydrochloric acid, hydrofluoric acid, phosphoric acid, mixed acid, ammonia water, and sodium hydroxide
Compressed and liquefied gases	Hydrogen, methane, silane, nitrogen, oxygen, argon, helium, ammonia, and chlorine

The Company has formulated a number of management policies and operation procedures, including the Chemicals Management Procedure, the Inspection and Registration System for Chemicals Coming In and Out of the Warehouse, the Hazardous Chemical Safety Management Regulations, and the Management Measures for Hazardous Chemicals, aiming at reducing the risks associated with chemical operations through standardized label management and in/out registration management.

The Company has established a Factory Chemical Review Committee and formulated the Chemical Substance Review Procedure, to comprehensively evaluate the environmental protection and safety qualifications as well as the risk prevention and control capabilities of chemical suppliers in advance. The Company employs the chemical substance management system to: manage the handling, storage, and disposal of chemicals; control the maximum storage quantity of chemicals; and dynamically monitor their use and consumption to ensure the safe use of chemicals and eliminates environmental pollution caused by chemicals.

Management Process of Hazardous Chemicals of Hua Hong Semiconductor



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Management for the Full Life Cycle of Products

The Company adopts the concept of circular economy, reduces the amount of raw materials used, continuously explores the recycling links of wastewater, energy, and waste, and lowers the impact of the full life cycle of products on the environment.

Environmental Management for the Full Life Cycle of Products of Hua Hong Semiconductor

 Warehousing of Raw Materials	 Product Production	 Product Transportation	 End-user Products	 Disposal
- Carry out hazardous substance examination and systematic control on raw materials - Review the qualification and environmental compliance of suppliers and require suppliers to sign the Environmental Protection Undertaking Letter	- Choose low-energy and high-efficiency production equipment to reduce the use of raw materials and energy - Treat production wastewater to meet standards and reuse it in production to increase the wastewater reuse rate - Optimize production technologies and pollution treatment processes to reduce the generation of wastewater and waste gas	- Use environmentally friendly packaging materials whenever possible - Recycle and reuse packaging materials - Optimize transportation routes to reduce repeated and no-load transportation	- Develop lowpower and highperformance chips to minimize environmental impact - Develop new products that are smaller in size but more efficient to reduce environmental pollution	- All wafers are tested for harmful substances to reduce their environmental impact when discarded - End-of-life products are disposed of as electronic waste by consumers through qualified institutions for harmless disposal, with some parts or metals being dismantled, refined, and recycled

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5. Caring for Employees and Promoting Harmonious Development

5.1 Rights and Interests of Employees

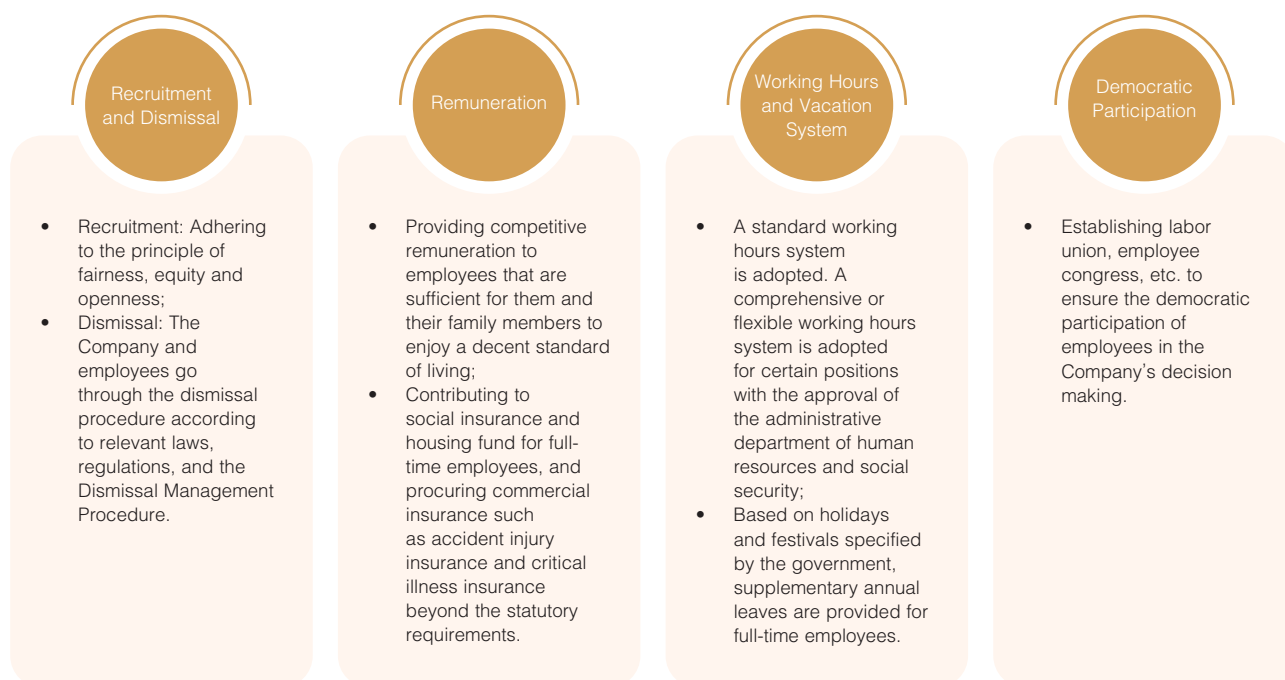
Employee Employment and Recruitment

The types of employee at Hua Hong Semiconductor include full-time contract workers, dispatched labor workers, and employees of other forms (like retired rehires and trainees). During the process of employment and recruitment, Hua Hong Semiconductor strictly adheres to the Labor Contract Law of the People's Republic of China and the Law of the People's Republic of China on the Protection of Women's Rights and Interests. We insist on fair employment, implement the principle of equal pay for equal work, and ensure that employees are not discriminated against or treated differently based on non-work factors including age, gender, place of birth, religious belief, marital status, or disability. We prohibit the employment of child labor or forced labor.

The Company has formulated relevant internal regulations, including the Employment Procedures, the Management Procedures for Trainees, and the Resignation Management Regulations. The Company strictly checks the identity information of employees and never employs adolescents under the age of 16. If such a situation is identified and verified, it will be immediately suspended and investigated for rectification, targeting at the omission of links around the recruitment review process, so that each process can be improved. During the Reporting Period, the Company has experienced no illegal or non-compliant incidents in connection with employee recruitment and dismissal, work hours and holidays, job promotion and equal opportunities, anti-discrimination and diversity, the employment of child labor or forced labor, nor had any litigation in connection with the aforesaid matters.

In order to better carry out the Company's talent reserve and cultivation, the Company launched a university campus summer internship program to provide internship opportunities for students to help them understand the Company's corporate culture, operation system, management model, etc. as soon as possible, for future employment and career preparation. The Company assigns senior professional staff to guide interns in all aspects of work, study and life, and provides certain remuneration for their work.

Employee Employment Norms of Hua Hong Semiconductor



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The Company firmly believes that a diverse and inclusive culture is a key factor in stimulating team vitality and promoting sustainable corporate development. In the recruitment process, the Company actively expands diversified recruitment channels to attract and accept outstanding talent with different cultural backgrounds, educational experiences, and technical expertise, while providing equal employment and development opportunities. Meanwhile, the Company attaches importance to diversity and equality in the workplace, ensuring that all employees can contribute their strengths in an environment that respects and supports differences.

Practices of Employee Diversity of Hua Hong Semiconductor

Gender Diversity

- Upholding the principle of “gender equality” in recruitment, employment, and promotion, the Company incorporates employee aptitude and performance assessments into the standards of employment and promotion.

Age Diversity

- Upholding the principle of “age equality”, the Company provides job and promotion opportunities for employees of different age groups.

Functional Diversity

- The Company offers various positions and levels in management, technology, and functional support, allowing employees to choose based on their own development plans.

Employee Composition of Hua Hong Semiconductor in 2024

Gender Distribution		
26%		74%
Female		Male
Age Distribution		
43%	55%	2%
<30 years old	30-50 years old	> 50 years old
Over 1,700 Persons Recruited in 2024, Including		
49%	35%	9%
Front-line production personnel	Engineering and technical personnel	R&D personnel

Employee Care

The Company creates a caring work environment for its employees, providing benefits in terms of office facilities, dining, accommodation, travel, etc., and fostering a warm, loving, and energetic work atmosphere.

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Employee Care System of Hua Hong Semiconductor

Office	- The ventilation system is installed in all offices and additional air purification devices added to improve office air quality. - The soft furnishings in employee lounges, such as refrigerators, TVs, and seats, are updated. - Greenery is increased within the factory premises, with the addition of fountains, decorative fishponds, and other landscaping features.
Diet	- There are staff canteens and coffee shops in all factories. - Establishing a food safety supervision team, adopting the mode of centralized purchasing of food materials and qualified supply chain management, and inviting employees and department representatives to carry out supervision. - Advocating healthy diet, constructing a healthy dietary system, scientific management of meals with nutritional balance and oil and salt control, effectively enhancing the rationality of employees' dietary structure, and realizing the dynamic balance between deliciousness and health.
Accommodation	- Construction of dormitory with private bathrooms, 24-hour supply of hot water, and WIFI network, etc. - The dormitory area has a library, computer room, snooker room, laundry, TV room, parcel locker, and other facilities.
Travel	- Providing all employees with travel allowance. - Providing free commuter bus services between the park, subway stations, and the dormitory.
Culture and sports	- Setting up various cultural and sports associations, including calligraphy and painting association, gardening association, tea art association, fitness association, cycling association, etc., which can be signed up for participation by all employees. - Organizing all kinds of competitions, including basketball, football, swimming, dry land curling, tennis, badminton, table tennis, etc., which can be signed up for participation by all employees.
Extending solicitude	- Celebrations and comfort activities are held during major holidays. - Providing holiday and birthday gift packages for employees, so that employees can feel the warmth and care of the Company.
Health checkup	- Offering annual health check-ups for all employees, including screenings for multiple types of cancer, and establishing employee health records for systematic tracking of changes in employee health status. - Setting up a "Health Protection Column", carrying out health special lectures, publish popular science articles to advocate healthy lifestyle. - Provide mutual aid insurance for employees every year, improve the level of medical insurance for employees, and reduce the medical burden of employees.

The Company complies with laws and regulations, such as the Measures for Labor Protection of Female Employees in Shanghai and the Measures for Urban Maternity Insurance in Shanghai, and has concluded a Collective Contract for the Protection of Female Employees, to safeguard the legitimate rights and interests of female employees. Mammy Huts are established across all factories, and the Labor Union allocates funds annually to support the development of female employees, planning festival welfare for International Women's Day each year.

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Employee Care Actions of Hua Hong Semiconductor in 2024

Action Name	Type and Effect
Expanding service areas	The Volunteer Service Corps is established to deepen volunteer joint construction activities in each factory area and to create a safe and healthy work environment, thereby continuously enhancing employees' sense of happiness, satisfaction and security.
Carrying out people-centric engagement programs	Bases in Shanghai and Wuxi have built new employee dormitories, and organized a rationalization proposal activity titled "Building Good Living Spaces, and Creating a Satisfactory Environment", combining contemporary employee needs to build leisure facilities.

Employee Communication

Employee communication is an important part of corporate sustainability. The Company is committed to establishing transparent and open communication mechanisms, to ensure that employees can fully express their opinions and participate in the Company's decision-making. The Company has formulated the Employee Communication Rules and encouraged multi-way communication between the Company and employees, between officers and subordinate officers, and among employees.

The Company listens to employees' voices and responds to their concerns in a timely manner by holding regular employee representative meetings, employee joint meetings, communication meetings and other forms. In addition, the Company allows employees to fully participate in the decision-making of the Company, promotes communication between management and employees, safeguards the legitimate rights and interests of employees and creates a harmonious working atmosphere. Through multiple forms and channels, such as conducting dynamic research on employees' thoughts, a rationalization suggestion platform and departmental communication meetings, the Company helps itself understand employees' needs, actively responding to reasonable demands, and enhancing employees' sense of responsibility and identity.

In addition, the Company conducts employee satisfaction surveys from time to time to collect opinions and needs of various employees extensively through different forms, including forums for young employees, an OA online platform, and internal departmental communications. The results of the employee satisfaction survey in 2024 remained at around 90 points, reflecting the overall recognition and satisfaction of employees with the work of the Company. In response to specific opinions put forward by employees, the Company actively responded and solidly promoted comprehensive supporting projects in Shanghai and Wuxi through the practice of "I do practical things for the masses", addressed the needs of employees, and further enhanced the sense of belonging and happiness of employees.

5.2 Employee Health and Safety

The Company strictly adheres to laws and regulations, such as the Occupational Disease Prevention and Control Law of the People's Republic of China, the Work Safety Law of the People's Republic of China, and the Work Injury Insurance Regulations, in its daily production and operation. To strengthen work safety efforts and establish a sound safety production responsibility system, the Company has formulated management procedures, including Implementation Measures for the Safety Production Responsibility System of the Safety Production Committee, the Environmental, Occupational Health, and Safety Management Manual, the Safety Production Expense Management Procedure, and the Work-Related Injury Management Method, in accordance with the requirements of the ISO45001:2018 Occupational Health and Safety Management System, thus providing employees with a good working environment and occupational health guarantees.

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Governance

The Company follows the principle of “one position with one responsibility” for the primary responsibility system of safety production, while improving the overall safety production responsibility system for all employees. The Company has established a Safety Production Committee, adhering to the principles of “strictly implementing primary responsibilities, strengthening direct supervision by departments, consolidating on-site supervision within the factory area, and reinforcing comprehensive safety supervision” and continuously promoting the standardization, systematization, and intelligent implementation of safety work.

The Safety Production Committee sets management objectives for safety production and occupational disease prevention and control, links safety management objectives with the safety production responsibility system, and regularly reviews the achievement of these objectives.

Strategy

The Company formulates an occupational health and safety management policy, strictly follows safety, health, and environmental protection laws and regulations, creates a safe, healthy, and comfortable working environment for its employees, and relentlessly pursues the ultimate goal of “zero disaster” and sustainable operation, aiming to become a model global corporate citizen.

The Company continuously promotes special rectification actions for safer production, covering aspects such as hazardous chemicals, engineering construction, fire protection, gas, electricity, and special equipment. During the Reporting Period, the Company held 4 safety committee meetings to deploy key tasks of the Three-Year Action Plan for Fundamentally Improving Safety Production. The management review meeting annually reviews the implementation of the occupational health and safety policy, updates legal and regulatory requirements, and deploys implementation plans.

Impacts, Risks, and Opportunities Management

The Company has formulated the Safety Production Expense Management Procedure, determines the budget at the beginning of the year, reduces risks and hazards in key areas and critical facilities, actively carries out safety technology improvement programs, and reports the progress of these programs on a weekly basis. The Company routinely identifies safety hazards, establishes emergency response infrastructure and teams, adopts a digital safety management system, clarifies work injury identification and handling methods, and establishes a complete safety production management system.

Safety Risk Identification and Management System of Hua Hong Semiconductor

Management Processes	Management Measures
Hazard identification	Grid-based Hazard Inspection: By adopting risk classification control and hazard identification, we have deeply promoted grid-based hazard screening. In 2024, we conducted a total of 410 supervision inspections and checked and corrected 16,139 hidden dangers. Special Inspection for Major Programs: We have strengthened safety management for major construction projects, implemented special actions for hazard identification and rectification, conducted safety inspections specifically for high-risk engineering tasks, adhered to twice-daily joint inspections, and successfully completed the “three simultaneous” phased work for major construction projects.
Emergency response	Fire Protection System Facilities: A comprehensive indoor and outdoor fire hydrant system, automatic fire extinguishing system, automatic fire alarm system, smoke exhaust system, fire emergency broadcasting system, and evacuation indication and emergency lighting system have been established.

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Management Processes	Management Measures
Digital management	Alarm Monitoring System: The clean room is equipped with alarm systems for toxic, harmful, and flammable gases, a liquid leak detection system, and a very early smoke detection alarm system. Emergency Response Team Support: An Emergency Response Team (ERT) has been established to conduct professional skills training. Emergency response plans tailored to different types of disasters have been developed. Specialized disposal drills are organized regularly, and issues identified are promptly rectified. Digital Systems: Safety management systems such as a special equipment registration system and an infrared inspection record system for equipment have been established using data records to ensure the implementation of rectification measures and standardize procedures.
Work-related injury handling	In accordance with the Work-Related Injury Management Regulations of the Company, when a work-related injury occurs, the department involved must report the incident to the safety department within 24 hours. The safety department shall then organize an investigation into it and convene a work-related injury seminar together with the Human Resources Department, the Labor Union, and the Medical Clinic under the Comprehensive Management Department and submit the application for work-related injury recognition to the social insurance administrative department.

Furthermore, to extend the development of our safety culture and enhance the safety awareness of all employees, the Company has launched thematic activities, such as “Safety for Everyone, and Emergency Response by All – Keeping Lifelines Open” for Safety Production Month and “Fire Safety for All, and Life Above All Else” for Fire Prevention Awareness Month. Through various safety promotion and exhibition activities, the Company aims to effectively improve employees’ safety awareness and knowledge and promote the fulfillment of safety commitments and responsibilities by all staff.

Safety Production Culture Creation Action of Hua Hong Semiconductor in 2024

Action Name	Content and Effect
Strengthening safety training	- Safety education on various safety topics and for all types of key personnel were conducted. In 2024, a total of 420 safety trainings were conducted, with a total of 60,458 person-times of employee participation and 27,789 person-times of contractor participation. - The participation rate in terms of safety education and training reached 100%, and the certification rate for personnel holding required certificates reached 100%.
Enhancing emergency response capability	- The expert review and filing of production safety emergency response plans have been completed, normative documents for establishing and improving rapid emergency response procedures against abnormalities at the production base have been put in place, and the goals of weekly training, quarterly drills, and annual competitions have been achieved. - Standardized training and actual combat drills for emergency rescue teams (ERT) were strengthened. In 2024, a total of 442 various emergency plan drills and 38 random ERT drills were organized, and more than 45,000 person-times of various skills training were carried out. - During the “Fire Prevention Month”, fire safety knowledge exhibition boards were displayed, and fire skills training as well as evacuation drills were conducted for all staffs, to enhance overall safety awareness and emergency response capabilities.

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Safety technical improvement assessment

- An evaluation activity for excellent safety technical transformation projects covering six dimensions was conducted. In 2024, a total of 52 safety technical improvements were implemented and 13 outstanding safety technical improvements were selected in the previous year.

The Company regularly identifies and evaluates safety risks. During the Reporting Period, 3 work-related injuries occurred in the Company, and targeted protective education has been strengthened in this regard, including safety training for all personnel in the affected departments and on-site engineering control measures.

Certification Status of Occupational Health and Safety Management System of Hua Hong Semiconductor

Entity	Certification Name	Validity Period of Certification
Shanghai Production Base	ISO45001: 2018 Occupational Health and Safety Management System	18 April 2027
Wuxi Production Base	ISO45001: 2018 Occupational Health and Safety Management System	18 April 2027

In order to prevent, control and eliminate occupational hazards and protect employees' occupational health and rights, the Company has established a sound occupational health management system. In terms of occupational disease prevention and control, the Company regularly identifies and monitors occupational hazard factors, informs employees of these factors upon their entry, provides consultation on personal protective equipment, formulates standards for protective equipment, regularly organizes occupational hazard post education and training, and supervises the correct use of protective equipment. The positions involving occupational hazard factors in the Company mainly include ion injection, diffusion, etching, chemical mechanical grinding, power gasification and other equipment operation positions.

Occupational Disease Hazard Prevention and Control System of Hua Hong Semiconductor

Job Positions with Occupational Hazards	Occupational Hazard Factors	Countermeasures	Occupational Health Monitoring
• Ion implantation, diffusion, etching, chemical mechanical grinding, power gasification, and other positions involving equipment operation.	• Fluorine and its inorganic compounds, hydrofluoric acid, hydrochloric acid, nitric acid, sulfuric acid, phosphoric acid, ammonia water, hydrogen peroxide, arsenic and its compounds, phosphorus and its compounds, isopropanol, etc.	• The equipment in the clean room is automatically operated in a closed space and is equipped with a closed process exhaust system; • Provide employees with personal protective articles and emergency response devices, regularly checking such articles and devices to ensure their effectiveness.	• Annual occupational health monitoring and assessment of current occupational hazards; • Strict pre-employment, on-the-job, and post-employment medical examinations for 100% employees at positions exposed to occupational disease hazards.

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Metrics and Targets

The Company has been certified with the Occupational Health and Safety Management System ISO 45001: 2018. All employees have signed the Safety Responsibility Letter with a 100% participation rate, and all contractors have signed the Letter of Commitment on Safety and Environmental Protection with a 100% commitment rate, so as to make every department and every employee take their respective responsibility. In 2024, the total investment in safety production was RMB34,162,300.

Occupational Health and Safety Metrics and Targets of Hua Hong Semiconductor in 2024

Metric	Unit	2024 Performance	2024 Target
ISO 45001 certification coverage	%	100	100
Employee signing rate of the Safety Responsibility Letter	%	100	100
Contractor signing rate of the Letter of Commitment on Safety and Environmental Protection	%	100	100
Annual number of work-related fatal accidents	Times	0	0
Annual number of occupational disease cases	Times	0	0

5.3 Employee Development and Training

Employee Training

The Company has established a sound employee education and training system, which is constantly adjusted based on strategic development and employee needs. In 2024, the Company updated personnel rules and regulations, such as Education and Training Procedures, to meet the all-round and multi-level training needs of employees in different positions.

The Company has complete training facilities, equipped with dedicated training classrooms and training equipment comma and an online platform for learning and sharing, including online training enrollment management, training materials and job question banks, multimedia learning courseware, etc. In addition, when necessary, the Company utilizes external resources to provide basic guarantees for employees' learning and development.

In 2024, the Company refined the classic programs of the special training camp for new college students, concentrated and updated the training courses, further stimulated the subjective initiative of learning through innovative forms, helped new college students to quickly integrate into the Company's and meet the challenges of future work with full enthusiasm.

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Employee Grading Training System of Hua Hong Semiconductor

Trainees	Training Contents	2024 Performance
Grassroots managers	Role recognition, self-management, managing others, work management	- Training hours per capita 137.30 hours - Employee training coverage 100%
Front-line managers	Develop management skills of front-line shift and team leaders, cultivate a front-line management team with high quality and high business ability, and lay a solid foundation for the Company's management	
Newly-employed university students	Career quality, corporate culture, introduction to special skills, and other courses	
Front-line employees	Courses about theory and practical training of the semiconductor manufacturing module	

The Company provides new employees with growth mentors, systematically imparting professional knowledge and practical experience to help them quickly adapt to the working environment and be competent at their jobs. The Company provides a variety of training courses covering job skills, professional technology, management ability, etc. We are committed to building a knowledge-based, skilled and innovative workforce, encouraging employees to carry out scientific research and innovation activities based on their positions, thereby enhancing the core competitiveness of the Company.

Employee Skill Enhancement Action by Hua Hong Semiconductor in 2024

Action Name	Actions and Achievements
Innovation studio	Based on 15 municipal and group-level employee innovation studios, the Company has established a platform for mass labor competitions in various forms and channels, hosting employee scientific paper contests, front-line employee skill competitions, energy-saving and emission-reduction groups, and other labor competitions, covering all departments and positions. Within the year, 105 papers were received, 178 patents were awarded, and 111 energy-saving and emission-reduction projects were completed.
Youth vanguard team	The Company launched company-wide youth vanguard team creation activities with the themes such as "Youth Taking Responsibility on 520, and Striving to Exceed by 100 Points", "Daring to Take on Responsibilities as Chip Youth, and Contributing to Hua Hong's High-Quality Development", and "Youth Setting Sail Forever Forward, and Assisting Hua Hong on Its New Journey". Over the past five years, a total of 147 company-level youth vanguard teams have been established, focusing on innovation, efficiency, and excellence in process development, flexible production, cost reduction, and supply chain safety.

Career Development

The Company is committed to building a clear and transparent career development path for employees, conducting regular performance evaluations and giving feedback. According to the standard practice of the industry, we have set up three professional development paths, including management, technology and function support, with corresponding job training, which not only supports employees to develop in a single sequence but also encourages conversion from the technical path to the management path, to take advantage of personal capabilities and promote professional aspirations.

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To ensure a high-quality talent pool, the Company enhances employees' comprehensive capabilities through various initiatives, including specialized training, technical exchanges, and lectures, to constantly optimize the talent cultivation mechanism. The Company has formulated the Measures for Academic Education Subsidy, which supports employees in pursuing on-the-job continuing education, enriching their professional knowledge and strengthening the Company's overall competitiveness.

5.4 Community and Public Welfare

The Company actively engages in social welfare, dedicating itself to contributing in various fields such as community service, education, and rural revitalization. It spreads love and warmth, establishing itself as a model of corporate responsibility.

Main Actions and Achievements of Community and Public Benefit of Hua Hong Semiconductor in 2024

Action Name	Action Content and Achievements
Volunteer Service	- Employee volunteers have been visiting the Meixin Nursing Home for 19 consecutive years, to carry out volunteer service, organizing activities such as singing, dancing, and doing handicrafts. They put up festive couplets at the nursing home during the Spring Festival, delivered zongzi and mooncakes on traditional festivals, and provided heat relief supplies in summer. - Employees donated blood 166 times.
Community Co-building	- During the Zhangjiang Science Hall Science Festival themed "Enlightening with Light, Exploring with Children (以光育光, 探索'童'行)", the Company organized a total of five sessions of integrated circuit-themed science popularization activities. - Through the fun course "Turning Sand into Gold", over 500 children were educated about the chip manufacturing process and the application of chips in daily life. - The Company has been providing the "First Aid Course for Accidents" in the community where the headquarters operates for seven consecutive years, covering basic first aid knowledge and cardiopulmonary resuscitation, to help residents master skills for handling emergencies.
One to-one Targeted Poverty Alleviation Program	- The Company launched a one to-one targeted poverty alleviation program with Yangbi Yi Autonomous Prefecture in Yunnan, purchasing local agricultural products, such as walnuts and tea, of approximately RMB1.2 million in aggregate. - The Company launched a one to-one targeted poverty alleviation program with Youqiao Village in Chongming District, Shanghai, donating RMB50,000.

6. Innovation-driven Development to Ensure Quality and Safety

6.1 R&D Innovation and Intellectual Property Protection

R&D Innovation

Relying on the layout advantage of "8-inch + 12-inch", the Company is committed to continuous innovation of specialized process technologies and building core competitiveness around embedded/stand-alone nonvolatile memory, power discrete, analog and power management, logic and RF and other specialized process platforms. Adhering to the advanced "Specialty IC + Power Discrete" dual-engine strategy, the Company is focused on high-speed penetration into emerging markets such as communications, new energy, Internet of Things, and automotive electronics.

Guided by the vision of "Continuous Innovation to Create 'Chip' Dreams for Global Customers", the Company strictly adheres to laws and regulations, such as the Law of the People's Republic of China on Scientific and Technological Progress and relevant requirements. It is committed to aligning with specialized technical processes, continuously enhancing its R&D innovation capabilities.

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Governance

The Company has established a governance structure comprising the “Board of Directors – Management – R&D Department”. The Board of Directors and Management are responsible for the Company’s overall R&D strategy and ensuring that the direction of R&D is consistent with the Company’s long-term goals. Management is responsible for implementing the Board of Directors’ decisions, formulating specific R&D plans and technology roadmaps, and coordinating resources to drive the innovation process. The R&D Department, as the core execution team, is responsible for day-to-day technology research, product development and innovation, to ensure continuous breakthroughs and improvements in new technologies and products.

Strategy

The Company recognizes the importance of continuously strengthening R&D innovation to refine manufacturing processes and continues to provide excellent products and services to customers around the world. When making strategic decisions on R&D innovation, the Company thoroughly evaluates the impact, risks and opportunities of innovation on product promotion. For example, the current digitalization trend is gradually enhancing the collaboration and influence between automotive manufacturers and chip designers in the supply chain. Additionally, the Company improves its resilience to risks by sensing market trends and enhancing resource utilization efficiency.

Impacts, Risks, and Opportunities Management

R&D Innovation Management System of Hua Hong Semiconductor

Management Process	Management Measures
Identification and Assessment	- Continuously monitor market changes and technological trends to ensure the technology platform adapts to evolving needs. - Regularly review internal resources and technical capabilities to ensure support for new product R&D and production. - Identify potential risk points, such as technical barriers, market acceptance, and regulatory changes, and assess their likelihood and financial impact.
Management and Optimization	- Reduce risk levels by improving processes and increasing R&D investment. - Increase investment in potential chip technology and promote research and development of relevant technologies. - Establish a flexible and efficient decision-making process to quickly respond to opportunities arising from market changes and technological advancements.

The Company has established broad product lines across multiple key technology platforms, with end-use applications covering diverse product needs, including consumer electronics, industrial control, automotive electronics, Internet of Things, communications and computers, and power management.

During the Reporting Period, the Company actively promoted the the beginning to end industrial ecological cooperation strategy and jointly promoted the sustainable development of the chip industry with its partners. Adhering to the development concept of “openness, innovation and cooperation”, the Company actively cooperates with upstream and downstream enterprises in the industrial chain. In the field of automotive electronics, with more than 20 years of successful experience in mass production of automotive electronic chips, the Company helped automotive products pass the AEC-Q100 Grade0 verification, and improved a zero-defect management mode for automotive electronics, providing comprehensive high-quality chip processing services for automotive electronic application systems.

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R&D Innovation Indicators and Targets of Hua Hong Semiconductor in 2024

Indicators	Unit	2023 Performance	2024 Performance
R&D Investment	RMB10,000	147,858	164,322
R&D Investment as % of Revenue	%	9.11	11.13
Number of R&D Personnel	Person	1,285	1,427
R&D Personnel as % of Total Employees	%	18.72	19.06
Honors and Awards	- National Science and Technology Progress Award (Second Prize) 35th Shanghai Outstanding Invention Gold and Silver Award (2024) - Shanghai Outstanding Academician Expert Workstation (2024) - Shanghai Employee Outstanding Innovation Achievement Award		

Intellectual Property Protection

The Company strictly adheres to laws and regulations, such as the Patent Law of the People's Republic of China, the Copyright Law of the People's Republic of China, and the Trademark Law of the People's Republic of China, as well as related requirements. It has established internal management systems, including the Intellectual Property Management System, to build a comprehensive and efficient intellectual property protection framework. This framework promotes technological innovation while standardizing intellectual property management.

The Company continuously optimizes and refines its intellectual property management mechanisms, striving to integrate intellectual property protection concepts into every business process. Each year, the Company sets goals for patent applications, actively safeguarding its own intellectual property. It commits to respecting and protecting the intellectual property rights of all organizations and individuals while implementing stringent protection measures for the product and technical information of upstream and downstream partners.

To mitigate the risks of intellectual property infringement, the Company standardizes and strengthens the assessment of customers' credibility and potential legal risks. Additionally, the Company has established technology licensing partnerships with multiple key technology providers to ensure that its products and services are developed on a legal and compliant basis.

Intellectual Property Protection Metrics and Targets of Hua Hong Semiconductor in 2024

Metric	Unit	2023 Performance	2024 Performance
Cumulative Number of Patent Applications	Piece	8,969	9,649
Cumulative Number of Patents Granted	Piece	4,427	4,644
Number of Patent Applications Completed	Piece	672	680
Number of Patents Approved	Piece	270	230

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6.2 Product and Service Safety and Quality

Product Quality and Safety

The Company consistently upholds high standards in product quality, adhering to the “zero-defect” philosophy and striving to deliver exceptional products and services to customers.

Governance

The Company complies with ISO9001, IATF16949 quality management system and related requirements. It has established internal management systems, including the Quality Manual, Code of Practice for Quality Objective Management, Non-conforming Product Control Procedure, and Supplier Management Regulations. These systems specify the responsibilities of the Quality and Reliability Assurance Department in ensuring product quality and safety, requiring regular reporting to management and the Board of Directors to promote comprehensive planning and implementation of quality management.

Strategy

The Company emphasizes the profound impact of quality management on product and business development, and treats quality as a core element of its corporate strategy. It has established a rigorous quality management system to identify and mitigate risk factors that may affect product quality, enhancing the Company’s adaptability to external environmental changes.

Impacts, Risks, and Opportunities Management

According to the Non-conforming Products Control Procedures, the Company has established a non-conforming products management rules covering the entire product cycle, specifying the handling specifications and applicable circumstances for each process, including incoming materials, production, testing, inspection, customer feedback, transportation and storage, etc. For identified non-conforming products, they are disposed in accordance with the relevant provisions of the regulations to prevent their unintended use.

Quality Management System Certifications of Hua Hong Semiconductor in 2024

Entity	Name	Expiration Date
Shanghai Base	ISO 9001 Quality Management System	17 May 2027
	IECQ QC 080000 Hazardous Substance Process Management System	11 December 2026
Wuxi Base	IATF 16949 Quality Management System for Automotive Industry	17 May 2027
	ISO 9001 Quality Management System	11 November 2025
	IECQ QC 080000 Hazardous Substance Process Management System	11 December 2026
	IATF 16949 Quality Management System for Automotive Industry	7 May 2026

During the Reporting Period, the Company implemented multiple measures to drive quality improvement, strengthen quality reviews, standardize supplier management, and conduct quality training, aiming to comprehensively enhance product quality and customer satisfaction.

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Product Quality Management Measures and Achievements of Hua Hong Semiconductor in 2024

Aspect	Specific Measures and Achievements
Driving Quality Improvement	- Conducted investigation and evaluation for five modules: R&D, production, manufacturing, testing, and services, and formulated improvement plans. - Proposed improvement solutions for each module and share our experiences in relation to the quality issues related to automotive-grade processes.
Strengthening Quality Reviews	- Established the Company Quality Objective (CQO) and conduct the review process. - Carried out activities such as the evaluation of the Quality Improvement Report Competition (Eight Disciplines Report) and high-quality project reviews.
Standardizing Supplier Management	- Established performance indicators for supplier management, including supplier complaint rates and scrap rates due to supplier raw material quality. - Established the Electronic Certificate of Analysis (ECO A) for suppliers and implemented Statistical Process Control (SPC) for key raw material parameters.
Conducting Quality Training	- The Shanghai and Wuxi bases conducted quality management and Zero Defects quality awareness training simultaneously, covering all employees, totaling 6,826 person-times. - Held quality month activities every year, including quality lectures and debates, supplier exchanges, information security weeks, and quality essay and calligraphy competitions, to promote quality culture and enhance employee quality awareness.

Product Quality Management Metrics and Achievements of Hua Hong Semiconductor in 2024

Metric	Unit	2023 Performance	2024 Performance	Target
Product Recalls	Time	0	0	0
CQO	%	93	84	80
Automotive Customer that Passed Audits	Time	14	33	–
Honors and Awards	- Shanghai Quality Technology Progress Award – Excellence Award: “Integration of 0.11-micron eFlash and BCD Automotive Process Platforms” - In 2024, the “Theme Activity of ‘Zero Defect of Characteristic Processes and High-quality Development of Hua Hong’” was recognized as the Quality Management Model (Organization) Award (Excellent Level) by the Shanghai Association for Quality			

Hazardous Substance Control

The Company places high importance on controlling hazardous substances in its products and follows international standards such as REACH, WEEE, RoHS and relevant requirements. It has established internal management systems, such as the Hazardous Substances Management Procedure, to build a comprehensive hazardous substance management system, ensuring product quality and safety. During the Reporting Period, the Company conducted internal research on the use of Per – and Polyfluoroalkyl Substances (PFAS). For raw materials containing PFAS, the Company collaborated with customers to evaluate the replacement of raw materials, ensuring compliance with overseas regulations.

During the Reporting Period, the Company implemented measures across three dimensions, namely source control, raw material testing and product certification, to ensure compliance and safety in the use of hazardous substances.

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Hazardous Substance Control Measures of Hua Hong Semiconductor in 2024

Aspect	Specific Measures
Source Control	• The Company incorporates hazardous substance control into the product R&D process to reduce the risk of using hazardous substances from the source.
Raw Material Testing	• Suppliers are required to provide the product test report for hazardous substances and sign the Letter of Commitment on Product and Environmental Protection. • We included hazardous substance compliance clauses in supplier audits and regularly conducted sampling tests to ensure the safety of raw materials.
Product Certification	• A third-party organization is entrusted to carry out product Testing and certification in accordance with the Directive on Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations (RoHS) and the Regulation concerning the Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) to ensure that our products meet the requirements of relevant certification standards.

Customer Relationship Management

Guided by the concept of “providing customers with more convenient and safe services”, the Company has strengthened its customer service management system, covering “customer communication – customer complaints – customer satisfaction surveys”. In adherence to applicable laws and regulation, including the Law of the People's Republic of China on the Protection of Consumer Rights and Interests and relevant requirements, the Company has formulated internal management systems, such as the Customer Complaint Management Procedure and the Customer Return Product Confirmation Procedure, to continuously elevate customer service standards and improve customer satisfaction.

To ensure prompt responses to customer feedback and concerns, the Company has outlined detailed standard procedures and response strategies for handling customer complaints in line with the Customer Complaint Management Procedure. Multiple customer complaint channels, including email, hotline, letter, and fax, confirming acceptance of customer complaints within 24 hours, providing 3D reports (problem prevention measures report) within 48 hours and submitting 8D reports (problem solving report) of common products within 14 calendar days, ensuring timely resolution of customer complaints. As of the end of the Reporting Period, the Company received a total of 2 customer complaints, achieving a 100% complaint resolution rate.

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Customer Complaint Handling Procedure of Hua Hong Semiconductor

Occurrence of a Complaint

Normally, communicate and confirm with the customer within 24 hours after the occurrence of a complaint and give a preliminary reply

Investigation into Failure Causes

Complaint cases requiring failure analysis shall be transferred to relevant departments for handling, corrective and preventive measures will be proposed according to the investigation results and a reply will be given to customers

Prevention and Correction

Implement corrective and preventive measures, sort out and analyze relevant information feedback by customers regularly and review the rectification effect

The Company consistently conducts customer satisfaction surveys. During the Reporting Period, the Company continued to enhance customer satisfaction by conducting customer satisfaction surveys and promoting quality improvement of its products and services, in accordance with the Customer Satisfaction Evaluation and Improvement Procedures. Management of the Company reviews the annual customer satisfaction survey results, conducts analysis and diagnostics, and implements corrective actions based on the results. The Company continuously monitors the effectiveness of these measures, while providing feedback to customers.

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Customer Satisfaction Survey Results of Hua Hong Semiconductor

2023	2024	Score change
8.97 points	8.89 points	↓ 0.08 point

6.3 Information Security and Privacy Protection

To continuously enhance information security and privacy protection, the Company complies with laws and regulations, such as the Cyber Security Law of the People's Republic of China, the Data Security Law of the People's Republic of China, the Cyber Security Management Regulations and the Regulation on the Administration of Commercial Cipher Codes, as well as other relevant requirements. During the Reporting Period, the Company revised management systems, including the Data Management Procedures and the Personal Information Management Procedures, in response to changes in external management regulations, to protect customer intellectual property, user privacy, and personal information security.

Governance

The Company continuously improves its Information Security Management System (ISMS) and has enhanced the management structure of "President – Information Security Committee – Information Security Working Group", to ensure the effective implementation and efficient operation of information security strategies. The Company has obtained ISO/IEC 27001:2022 Information Security Management System certification, with a validity period until February 2025. During the Reporting Period, the Company did not experience any complaints or incidents related to the infringement of customer privacy or loss of customer data.

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ISMS Structure of Hua Hong Semiconductor

President: As the top manager of ISMS, takes overall control of information management

Information Security Committee: It is composed of management representatives and department heads to study and approve the Company's supervision rules and regulations on information security and to supervise the duty performance of relevant departments

Information Security Working Group: With its members from the Information Technology Department, Quality and Reliability Assurance Department, Compliance Department, etc., it is responsible for the establishment and implementation of the ISMS

Strategy

The Company fully acknowledges the critical importance of information security and privacy protection. In response to the dual challenges of increasingly stringent regulatory requirements and rapid technological advancements, the Company has embedded information security into its overall development strategy. By proactively allocating resources and establishing a systematic management framework, the Company strives to identify and mitigate potential risks in advance, while capitalizing on strategic opportunities in the field of information security.

Impact, Risks, and Opportunities Management

During the Reporting Period, the Company enhanced its overall capabilities in information security and privacy protection through initiatives such as personal information access management, risk assessments and self-inspections, and the execution of attack-defence drills. The Company is dedicated to establishing a more comprehensive, efficient, and sustainable information security management system.

Information Security and Privacy Protection Management System of Hua Hong Semiconductor in 2024

Management Process	Management Measures
Information access	- Enhanced protection measures across all stages of personal information processing and upgraded the visitor registration system. - Revised the Personal Information Management Procedures to enforce privacy protection requirements.
Risk assessment	Strengthened the information security management system and revised the Information Security Risk Assessment Management Procedures to comply with data classification and grading requirements.
Risk self-inspection	Promoted special initiatives on cyber security, attaining a “Level 2 Networked Industrial Enterprise” designation from the Ministry of Industry and Information Technology.
Practical drills	Implemented cyber security protection reinforcement initiatives and participated in the “Cyber Security Enhancement with Industrial Empowerment” practical attack-defence drill organized by the Shanghai Municipal Commission of Economy and Informatization, receiving the “Outstanding Blue Team Enterprise” award.

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Metrics and Targets

The Company conducts quarterly training sessions on information security and privacy protection, monitors participant numbers, and reviews training outcomes. Training topics include system certification, information confidentiality, trade secret protection, and strategies to counter cyberattacks.

Information Security and Privacy Protection Training of Hua Hong Semiconductor in 2024

Theme	Coverage	Participants in 2023	Participants in 2024
ISMS Training – Q1	Shanghai & Wuxi Production Bases	8,233	6,683
ISMS Training – Q2		6,554	6,117
Introduction to the New Version of ISO 27001 System Standards and Information Security Awareness Training – Q3		6,476	7,346
Introduction to the New Version of ISO 27001 System Standards and Information Security Awareness Training – Q4		6,919	7,491
Confidentiality and Information Security Compliance Training		6,802	7,314
Specialized Training for Attack and Defense Exercises		–	7,525
ISO 27001 Standards and Audit Skills Training		–	99

6.4 Sustainable Supply Chain Management

Sustainable Supply Chain Management

The Company primarily relies on suppliers of silicon wafers, chemicals, and gases. The Company actively adheres to industry standards and norms, such as the Responsible Business Alliance (RBA). During the Reporting Period, the Company received the RBA “Platinum Award” certification.

Governance

The Company has established a supply chain task force composed of engineering departments from various bases, the Quality and Reliability Assurance Department, and the Procurement Department. The management team is responsible for setting strategic directions and overseeing execution, with regular reporting to the Board to ensure that supply chain policies align with the Company’s objectives. The supply chain task force is tasked with evaluating, selecting, and conducting regular audits of suppliers, establishing a supplier performance management system and compliance review mechanism, to ensure the comprehensive implementation of quality, environmental, and social responsibility requirements.

The Company has formulated management systems, such as the Supplier Risk Identification, Planning, and Control Management Procedure and the Social Responsibility Management Manual, to clarify the responsibilities and requirements of supply chain management, ensuring a scientific and rigorous process for supplier screening, qualification, and management.

Strategy

By establishing a systematic supply chain management process, the Company integrates multi-dimensional risks and opportunities related to the supply chain into its overall strategy, promoting the development of a green supply chain. Through regular identification and assessment of key influencing factors within the supply chain, the Company ensures that its management strategies can effectively respond to external environmental shocks and market changes.

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At the same time, the Company deeply integrates supply chain management with its strategic goals. Not only does it ensure business continuity through risk control, but it also actively explores the potential of the supply chain in cost optimization, technological collaboration, and localized development. This enables the synergistic advancement of supply chain management and the Company's development.

Overview of the Social Responsibility Requirements for Suppliers of Hua Hong Semiconductor

 Labor	- Prohibition of forced labour - Non-employment of minors	- Working hours - Wages and benefits	- Humane treatment - Non-discrimination/non-harassment - Freedom of association and collective bargaining
 Environmental Health and Safety	- Occupational safety - Emergency preparedness - Work injury and disease	- Industrial hygiene - Heavy work - Mechanical safety protection	Public health, food and accommodation
 Environmental protection	- Environmental licensing and reporting - Pollution prevention and resource conservation - Hazardous substances	- Solid waste - Exhaust emissions - Restricted materials	- Water management - Energy consumption and greenhouse gas emissions
 Business ethics	- Business integrity - No unfair advantage - Information disclosure	- Intellectual property rights - Fair dealing, advertising and competition - Identity protection and prohibition of retaliation	- Responsible mineral sourcing - Privacy
 Management system	- Management responsibilities and employee responsibilities - Legal and customer requirements - Risk assessment and risk management	- Improvement objectives - Training - Communication	- Audit and evaluation - Corrective action

Impacts, Risks, and Opportunities Management

The Company adheres to the Responsible Business Alliance (RBA) standards for social responsibility audits in the electronics industry. It clearly defines the basic requirements for suppliers in terms of quality, supply stability, labor management, environmental protection, and business ethics. The Company has established a management system that includes initial reviews before supplier access, annual evaluations, and regular training sessions. This system helps maintain stable cooperative relationships with high-performing suppliers.

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Impacts, Risks, and Opportunities Management System of Sustainable Supply Chain of Hua Hong Semiconductor

Management Process	Management Measures
Supplier access	During the supplier access phase, conduct initial reviews of performance in areas such as quality requirements, stable supply, labor management, environmental protection, and business ethics. Screen out suppliers with negative information or those that fail to meet requirements. Key suppliers must pass the initial review before being registered on the Approved Supplier List (ASL) to mitigate risks at the initial stage of the supply chain.
Supplier review	Conduct comprehensive risk assessments annually. In accordance with the Supplier Risk Identification, Planning, and Control Management Procedure, identify potential risk points in suppliers through third-party audits or audit questionnaires, and propose corrective actions for identified risks. If a supplier fails to implement effective corrections or commits major violations, the cooperative relationship will be terminated to ensure supply chain stability and compliance.
Performance evaluation	- Strictly control key performance, such as raw material quality and compliance with hazardous substance contents. - Conduct annual supplier performance evaluations and recognize outstanding suppliers with awards to incentivize high-performing suppliers.
Supplier dissemination	Regularly conduct dissemination and training covering multiple categories of suppliers. Align suppliers with environmental, ethical, and quality standards, provide technical support, and jointly explore potential opportunities for supply chain optimization and improvement.

The Company enforces strict anti-bribery and anti-corruption policies for suppliers, requiring all suppliers to sign a business ethics commitment letter. Through regular audits and performance evaluations, the Company ensures that their business practices align with its integrity and compliance standards.

The manufacturing process of semiconductor silicon wafers requires a significant amount of water resources and carries a high risk of pollution emissions. Failure to properly address pollution emissions and wastewater treatment during production could exert greater pressure on water resource regeneration. To address this, the Company has implemented water resource management initiatives and performance investigations for silicon wafer suppliers, including water-saving goals, water consumption, and water-saving measures, aiming to mitigate the impact of their production operations on water resources. The Company set a goal of “achieving 100% coverage of water resource investigations for major silicon wafer suppliers”, which was accomplished during the Reporting Period.

During the Reporting Period, the Company conducted 270 investigations on related enterprises and established contacts with over a hundred domestic suppliers, further optimizing the supply chain management system. This ensures the stability and efficiency of the supply chain, providing strong support for business development.

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Metrics and Targets of Sustainable Supply Chain Management of Hua Hong Semiconductor in 2024

Metric	Performance in 2023	Performance in 2024
Coverage Rate of Social Responsibility Review for Key Supplier	100%	100%
Number of Training Programs for Key Suppliers	20	20
Number of Newly Introduced Domestic Suppliers	28	18
Diversification rate of “8-inch + 12-inch” raw material suppliers	65.4%	76%
Diversification rate of “8-inch” raw material suppliers	72%	84%
Diversification rate of “8-inch” target materials and silicon wafers suppliers	> 90%	98%
Diversification rate of “12-inch” raw material suppliers	55%	65%

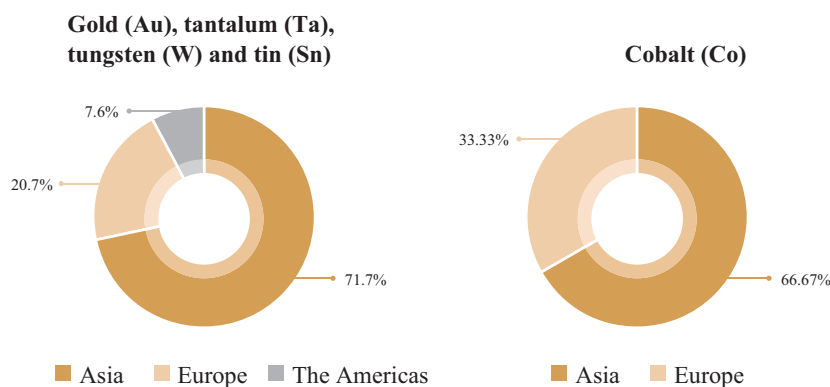
Responsible Minerals Management

“Conflict Minerals” refer to minerals mined in conflict zones that fund armed groups through illegal trade. These minerals are extracted through violent coercion of laborers, child labor, and damaging the environment and ecology. The trade of such minerals frequently leads to human rights violations, environmental damage, and social instability. According to the Dodd-Frank Wall Street Reform and Consumer Protection Act and research by international nongovernmental organizations, conflict minerals may enter the information, communication, and technology (ICT) industry and be used in electronic and electrical products, such as mobile phones and computers.

In the Company’s production process, minerals such as gold (Au), tantalum (Ta), tungsten (W), tin (Sn), and cobalt (Co) are used. To prevent the procurement and funding of mineral sources with negative social and environmental impacts, the Company has established the Conflict Minerals Management Policy to ensure that the minerals procured do not originate from conflict zones.

Over 71% of gold, tin, tantalum, and tungsten used in the Company’s production process are from Asia, with over 20% from Europe, and the remaining portion from the Americas. Two-thirds of the cobalt used comes from Asia, and one-third from Europe. As of the end of the Reporting Period, none of the gold, tantalum, tungsten, or tin used by the Company was from the regions with armed conflict.

Types and Sources of Minerals Used of Hua Hong Semiconductor



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In addition, the Company adheres to international standards such as the Responsible Minerals Initiative (RMI) and the Organization for Economic Co-operation and Development (OECD) Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas. It conducts traceability and due diligence on suppliers, requiring all suppliers to promise not to purchase conflict minerals in Conflict-Affected and High-Risk Areas. This ensures that the mineral sources in the supply chain comply with regulatory and ethical requirements.

Conflict Minerals Management Policy of Hua Hong Semiconductor

1. Suppliers whose raw materials contain gold (Au), tantalum (Ta), tungsten (W), and tin (Sn) are required to purchase materials according to the Responsible Minerals Policy, while suppliers whose raw materials contain cobalt (Co) are required to disclose the smelters of cobalt

The Company takes global social and environmental responsibility as its goal and follows green procurement principles and has promised to undertake the following social and environmental responsibilities in its metal supply chain

2. Suppliers are required to provide the declaration that they do not use gold (Au), tantalum (Ta), tungsten (W), tin (Sn), and cobalt (Co) from "conflict minerals" and we issue the Conflict Minerals Reporting Template (CMRT) and the Extended Minerals Reporting Template (EMRT) to suppliers

During the Reporting Period, the Company conducted questionnaire surveys on metal target materials suppliers, achieving a coverage rate of 100%, including both newly introduced and existing suppliers. The Company traced the sources of mineral raw materials used in its products to ensure that the origin information of each batch of minerals was clear and traceable. The traceability scope covered mines, smelters, and the entire supply chain. During the Reporting Period, the traceability results showed that 100% of the Company's suppliers do not use conflict minerals.

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7. Robust Governance and Adherence to Business Ethics

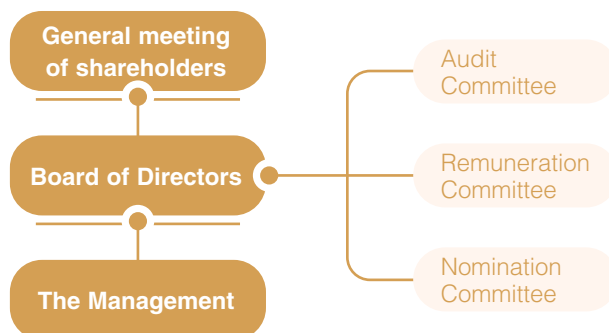
7.1 Corporate Governance

Corporate Governance

As a listed company on the Main Board of the Hong Kong Stock Exchange and the STAR Market of the Shanghai Stock Exchange, the Company strictly complies with applicable laws and regulations of the place of incorporation, the place of overseas listing and the PRC, such as the Companies Ordinance of Hong Kong, the Listing Rules of the Stock Exchange, the Securities Law of the People's Republic of China, and the STAR Market Listing Rules of the Shanghai Stock Exchange. It has established regulations such as the Articles of Association, the Rules of Procedure for Shareholders' General Meetings, and the Rules of Procedure for Board of Directors, forming a corporate governance structure and operational mechanism where the shareholders' general meeting, the board of directors, and its specialized committees work in coordination. This promotes scientific decision-making, standardized operations, and further enhances the Company's management level.

The Company's Board has established specialized committees, including the Audit Committee, Remuneration Committee, and Nomination Committee. Each committee performs its respective duties, overseeing the Board's responsibilities and supporting its scientific decision-making. The Company has appointed three Independent Non-Executive Directors to participate in decision-making and supervision, enhancing the objectivity and scientific rigor of the Board's decisions.

Corporate Governance Structure of Hua Hong Semiconductor



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As of the end of the Reporting Period, the Board of the Company consists of 9 members, including 1 female director. During the Reporting Period, the Company held 16 Board meetings and considered 58 proposals.

Information Disclosure

The Company strictly adheres to applicable laws and regulations and relevant requirements, such as the Administrative Measures for the Disclosure of Information of Listed Companies of the China Securities Regulatory Commission (CSRC), and has established relevant documents, such as the Information Disclosure Management System. It is committed to ensuring the authenticity, accuracy, timeliness, fairness, and completeness of information disclosure to meet the information needs of stakeholders. During the Reporting Period, the Company strictly complied with relevant information disclosure regulations, promptly reported significant matters, and continuously improved the quality and compliance of its information disclosure.

The Company regulates the shareholding behavior of directors and senior management by disclosing their shareholding plans. In August 2023, the Company released the Announcement Regarding the Plan for Additional Acquisition of A Shares of the Company by Certain Senior Management, explicitly mentioning the share increase plans of executive directors and senior management. The plan was fully implemented by 4 January 2024. Through centralized bidding, the Company's senior management collectively increased their holdings of the Company's A-shares by 53,618 shares, with a total amount of RMB2.3094 million. After the increase, the total shareholding ratio rose to 0.0275%.

Protection of Investors' Rights and Interests

The Company places great emphasis on protecting shareholders' rights and interests. By formulating reasonable profit distribution policies and dividend plans, it actively rewards shareholders and is committed to creating stable long-term returns for investors. The Company has established an investor relations management department to maintain proactive communication with stakeholders, address key concerns of investors, and ensure that their reasonable demands are properly handled.

Adhering to the principle of "equal treatment of all investors" and the requirement of "compliant information disclosure", the Company follows the guidance of "integrity, compliance, and interactive communication". It disseminates information about the Company's operations, finances, product technologies, and significant matters to the capital market through various channels, such as shareholders' general meetings, regular reports, direct hotline of investors, SSE e-Interaction, special email for investors, online and offline communication meetings, official website and WeChat account of the Company.

During the Reporting Period, the Company held 1 shareholder general meeting and considered 13 resolutions. All meetings were open to all shareholders, including small and medium-sized investors, effectively safeguarding their participation and right to know. Additionally, the Company organized 4 performance exchange meetings to engage in in-depth discussions with investors, enhancing transparency and trust.

7.2 Risk and Compliance Management

The Company places great importance on risk and compliance management, strictly adhering to the laws and regulations applicable to its operating locations. It closely monitors the promulgation, changes, and revisions of laws and regulations, identifies provisions relevant to the Company's operations and production, and promptly aligns internal management systems with external regulations to ensure compliance with the requirements of laws and regulations.

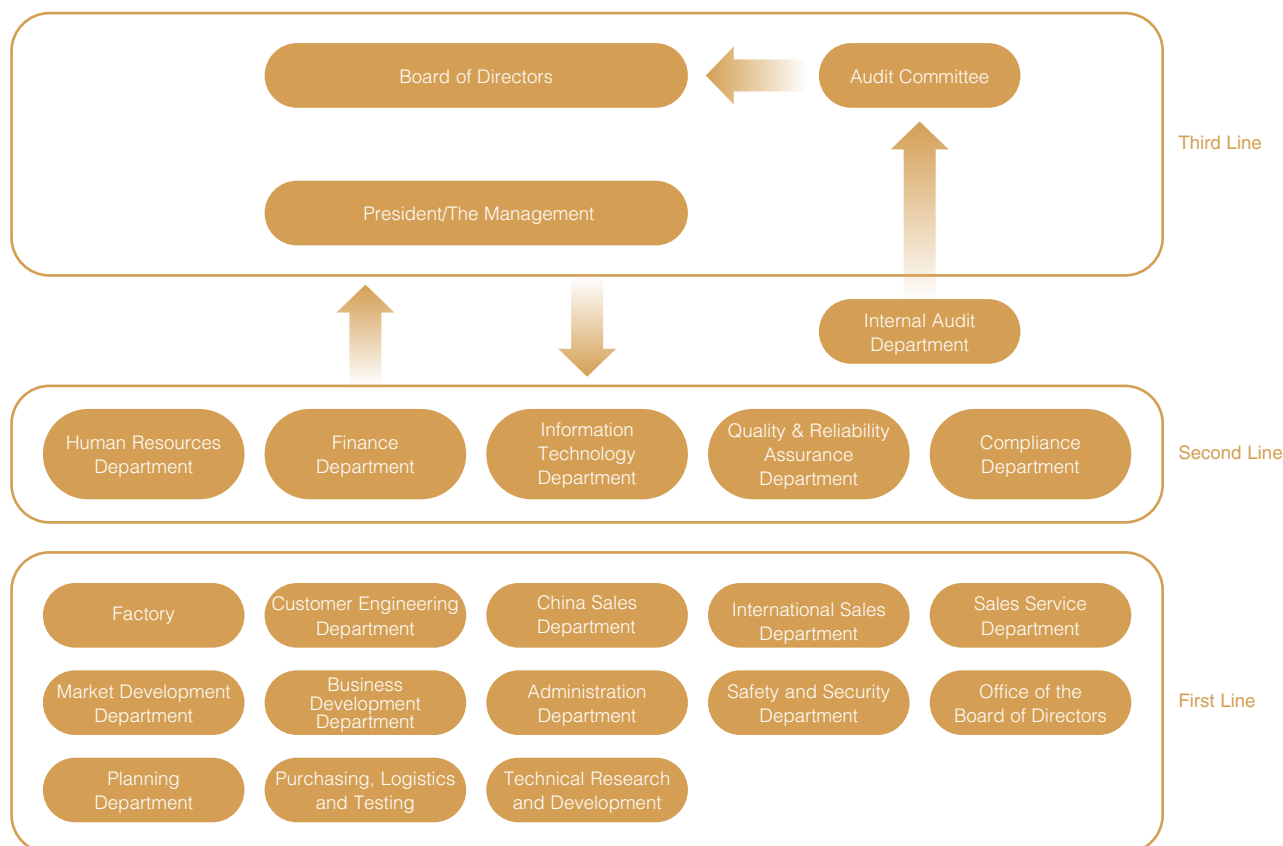
The Company has established the Rules about Comprehensive Control of Risks that covers all business processes. By regularly identifying and analyzing risk lists, it builds a long-term management mechanism, implements risk prevention measures across different levels and business lines, and enforces effective risk control measures in key areas.

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Drawing on the “Three-Line Model” by the Institute of Internal Auditors (IIA), the Company, under the authorization of the Audit Committee and management, has established a risk management system tailored to its characteristics. It clearly defines the division of responsibilities for addressing major risks, including the responsible leaders, leading departments, and their respective roles.

Hua Hong Semiconductor’s Three Lines of Defence for Risk Management in the Economic Sector

Organisation Governing Body (accountable to stakeholders, with supervisory responsibilities)



Functions of the First Line: to provide products/services to customers and manage risks;

Functions of the Second Line: to provide expertise, support, monitoring, and reasonable challenge on risk-related matters;

Functions of the Third Line: to provide independent and objective confirmation and advice on all matters related to the achievement of objectives.

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Based on the “Three-Line Model” of risk management, the Company adheres to the principle of combining comprehensiveness and importance, continuously optimizes its risk map, integrates and categorizes certain similar risk factors, revises and supplements examples of risks, and identifies 59 major risk factors. This enables comprehensive prevention and effective supervision of both internal and external risks.

During the Reporting Period, the Company conducted annual risk assessments at its Shanghai and Wuxi Production Bases. It designed risk survey questionnaires and systematically analyzed and identified key risk areas through one-on-one interviews, multi-department seminars, on-site research sampling, and other methods.

Risk Management System of Hua Hong Semiconductor

Management Process	Management Measures
Risk assessment	• We identify and analyze risk areas through methods such as questionnaires, one-on-one interviews, and multi-department seminars and score the risk areas to generate an annual risk management report.
Risk reporting	• We implement a quarterly risk communication and reporting mechanism, where each business unit collects and reports risk events within their respective areas on a quarterly basis or as needed.
Communication and training	• From time to time, we organize risk line management meetings, conduct risk training, and communicate risk issues across business units and at the company level.

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Risk Response Mechanism of Hua Hong Semiconductor

Risk Category	Risk Content	Strategy
Operation	Research and Development	Continuously improve the R&D project management system, comprehensively monitor the project initiation, implementation and post-evaluation of R&D, continuously improve the project manager's ability, and timely develop new technologies and products with commercial value
	Product Structure	Continuously promote technical product upgrading iteration, increase flexible production capacity, promote process interoperability among factories, and optimize product structure
	Talent Reserve and Development	Establish a talent resume database, develop talent inventory, continuously explore recruitment channels, optimize salary structure, improve supporting welfare policies, and comprehensively enhance the attractiveness of enterprises and employee satisfaction
	Supply Chain	Continuously develop diversified supply channels to ensure the stability of the supply chain; set safety stock values for each production material, regularly review the rationality of safety stock values, and timely adjust high-risk materials with market changes
	Information Security	Establish information security framework and management policy, implement information security risk assessment process every year, and continuously monitor all critical information through DLP data protection system to ensure that the best interests of the Company, shareholders, customers, suppliers and employees are protected
	Exchange Rate Control	Adhering to the concept of exchange rate risk neutrality, the Company reduces the exchange rate risk of existing US dollar debts through currency conversion, interest rate swaps and the adoption of exchange rate risk management products
Finance		
Environment and Safety	Security Inspection	Continuously and regularly carry out safety inspection focusing on hidden danger investigation and equipment fault diagnosis
	Occupational Health	The Company formulates environmental occupational health and safety objectives, indicators and scheme management tables according to the objectives, indicators and scheme management procedures of the HSE management system
	Environment	The Company identifies, evaluates and controls the planning and management procedures according to environmental factors, and makes a list of major environmental factors

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7.3 Business Ethics

Anti-commercial Bribery and Anti-corruption

The Company is consistently committed to fostering a corporate culture of integrity, honesty, and fairness. It strictly complies with laws and regulations, such as the Criminal Law of the People's Republic of China and the Interim Provisions on Prohibition of Commercial Bribery, and has established internal management systems, including the Social Responsibility Management Manual, the Undertaking System on Anti-Corruption and Business Ethics, and the Business Ethics Talk Measures. These ensure adherence to high standards of integrity in partnerships and daily operations.

The Company integrates business ethics into its overall governance framework. Management is responsible for overseeing the implementation and execution of this strategy, while functional departments, such as the Compliance Department, Human Resources Department, and Procurement Department, are specifically tasked with the management and supervision of business ethics.

Anti-corruption and Anti-bribery Mechanism of Hua Hong Semiconductor

All employees (including part-time employees), senior management, and Board members are required to abide by relevant laws and regulations and practice and behave with integrity, diligence, and self-discipline. Corruption and bribery in all forms are prohibited, including:

- prohibiting commercial bribery and maintaining fair competition order;
- prohibiting seeking for illegitimate benefits through taking advantage of influence in position and work;
- never violating financial management and operation regulations, or seeking personal gain in the name of the Company;
- prohibiting practicing fraud to undermine the legitimate interests and reputation of the Company;
- prohibiting any receipt of presents, cash gifts, securities and monetary items that may cause a negative influence on work.

The Company has established a systematic management system for business ethics to ensure that all operation activities of the Company always comply with high standards of business ethics. The Company regularly conducts comprehensive assessments of business ethics and anti-corruption efforts, covering all business operations. The Company not only assesses the effectiveness of anti-corruption mechanisms, but also identifies potential integrity risks and proposes improvement measures. In addition, the Company has established a regular verification mechanism for business ethics to ensure that ethical policies are consistent with business developments and industry norms.

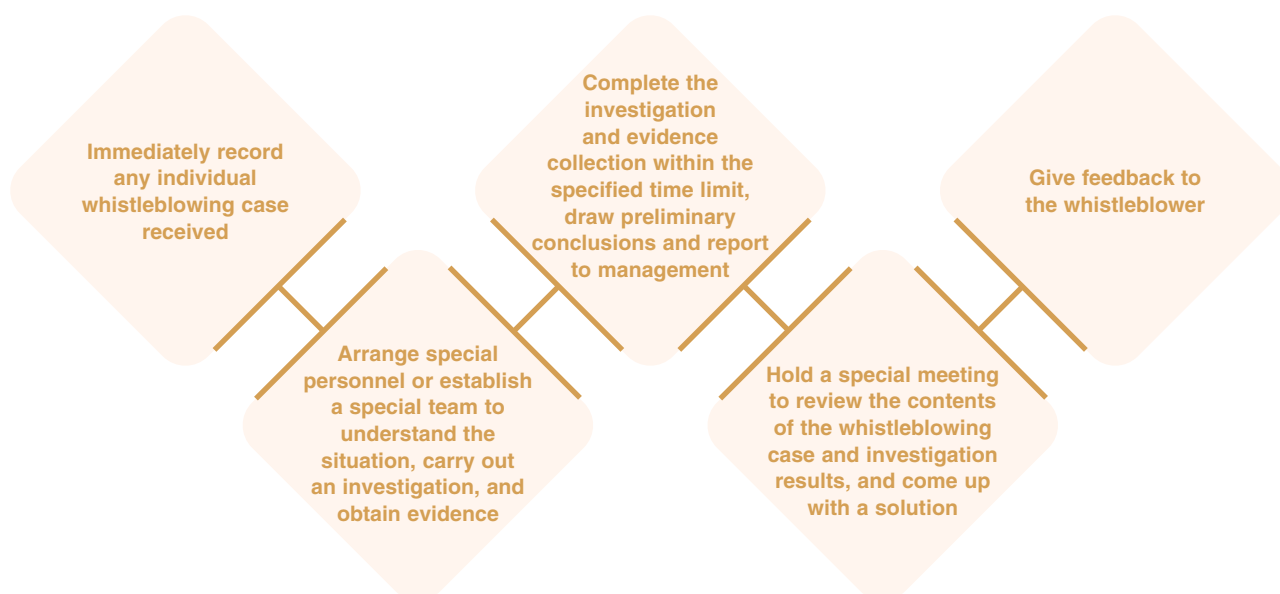
The Company is committed to creating a clean business ethics atmosphere. It conducts Responsible Business Alliance (RBA) Code of Conduct training for new employees, emphasizing clean business and other ethical business behaviors; and organizes anti-corruption and business ethics training, covering all employees in key positions, to promote the in-depth development of a culture of integrity and create a healthy work environment. During the Reporting Period, signing Ceremonies for the Undertaking on Business Ethics were held in Shanghai and Wuxi, targeting the department heads and above, aiming to reinforce the commitment to ethical conduct and further enhance the transparent atmosphere within the Company.

The Company encourages employees to actively participate in supervision and whistle-blowing to maintain a clean working environment, and provides multiple whistle-blowing channels, such as email, telephone and mailbox, and employees can choose to report in real name or anonymously. The Company always maintains an open attitude to ensure timely handling of reported information, so as to promptly identify and solve problems and prevent and control risks.

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To safeguard the fairness and transparency of whistle-blowing, the Company has established a sound business ethics supervision and whistle-blowing handling mechanism. All reported cases are investigated in detail and the results are reported to the whistleblower after consideration. With regard to whistleblower protection, the Company ensures that their information is kept strictly confidential and prevents whistleblowers from being dismissed, demoted, suspended or subjected to any form of retaliation for taking a lawful approach to whistle-blowing.

Hua Hong Semiconductor's Business Ethics Supervision and Whistle-blowing Case Handling Procedures



As of the end of the Reporting Period, the Company had no commercial bribery or corruption related incidents and litigation cases.

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Hua Hong Semiconductor’s 2024 Business Ethics Metrics and Targets

2024 Target	Achievement of 2024 Target
All suppliers have signed the Anti-Commercial Bribery Commitment Letter	100%
All employees in critical positions have signed the Integrity Commitment Letter	100%

Anti-Unfair Competition

Hua Hong Semiconductor strictly complies with the Anti-Unfair Competition Law of the People’s Republic of China and other laws and regulations and relevant requirements, explicitly prohibits unfair competitive behaviors, such as false advertising, trade secret infringement, and abuse of market dominance. The Company has established a relatively comprehensive governance structure and internal supervision and management procedures, implemented risk assessment and compliance review processes, and ensure timely identification and correction of any unfair competitive behaviors. Meanwhile, the Company conducts regular training for its employees to strengthen their legal compliance awareness, thereby preventing and addressing unfair competitive behaviors.

The Company has clearly defined the responsibilities of each department to ensure effective identification of and response to unfair competition risks. At the same time, through strengthened analysis of the market and competitive environment, the Company conducts regular assessments of potential risks and opportunities and formulates corresponding strategies to make adjustments.

As of the end of the Reporting Period, the Company has not been subject to any unfair competition lawsuits or administrative penalties, nor had it violated any laws or regulations regarding the use of product labels and promotional content.

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8. Appendix

8.1 ESG Data and Notes

Environment¹

Performance Indicators	Unit	2022	2023	2024
Emissions				
Total exhaust air emissions	10,000 m ³	2,391,024	2,747,929	2,478,838
Nitrogen oxide (NO _x) emissions	Kg	32,650	33,719	35,333
Sulfur dioxide (SO ₂) emissions	Kg	3,546	5,252	4,013
Total wastewater discharge	10,000 m ³	832	898	970
Total Hazardous waste produced ²	Ton	20,385	20,797	27,601
Hazardous waste produced per unit product	Kg/8-inch wafers	4.88	5.29	6.07
Total Non-hazardous waste produced ²	Ton	9,864	9,269	10,134
Non-hazardous waste produced per unit product	Kg/8-inch wafers	2.36	2.36	2.23
Use of Resources				
Integrated energy consumed ³	MWh	1,065,002	1,168,170	1,272,331
Integrated energy consumed per unit product	MWh/8-inch wafers	0.25	0.30	0.28
Of which: Electricity purchased ¹	MWh	954,667	1,030,352	1,104,829
Natural gas consumed ¹	m ³	10,530,287	11,048,312	11,630,543
Gasoline consumed ¹	L	–	43,949	43,923
Diesel consumed ¹	L	–	15,911	21,244
Heat purchased ¹	GJ	–	134,940	141,125
Total water consumed ⁴	m ³	18,010,226	19,076,734	21,295,153
Of which: Water from municipal water supply	m ³	10,284,063	11,127,266	12,560,168
Wastewater reused	m ³	7,726,163	7,949,468	8,734,985
Water consumed per unit product ⁵	m ³ /8-inch wafers	2.46	2.83	2.76
Recycled/reused water	m ³	107,163,560	126,544,660	148,470,458
Total packaging materials used for the shipment of finished products	Ton	313.32	336.92	420.32
Packaging materials used for the shipment of per unit finished product	Kg/8-inch wafers	0.07	0.09	0.09
Recycled packaging materials used for the shipment of finished products	Ton	54.90	66.33	84.71
GHG Emission				
GHG emissions ⁶	tCO ₂ e	497,938	537,070	562,487
Of which: Direct GHG emissions	tCO ₂ e	24,877	22,881	25,331
Indirect GHG emissions ⁷	tCO ₂ e	473,060	514,189	537,156
GHG emissions per unit product	tCO ₂ e/8-inch wafers	0.12	0.14	0.12

Notes:

1. The expansion project of the Wuxi Production Base was put into operation in 2023, and the production capacity was expanded in 2024. Therefore, integrated energy consumed, total wastewater discharge, waste produced increased in 2024, but the intensity performance indicators of most unit products have decreased.
2. Hazardous waste mainly includes waste isopropanol, waste acid, waste oil, waste packaging materials, etc. Hazardous waste mainly includes waste sludge and domestic garbage.
3. Integrated energy consumption included electricity purchased, natural gas, gasoline, diesel and heat purchased.
4. Total water consumption = water consumption from municipal water supply + wastewater reuse.

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5. In calculation, water consumed per unit product only includes water from the municipal water supply.
6. GHG emissions are calculated in accordance with the GB/T 32150 General Guideline for Calculation and Reporting of GHG Emissions from Industrial Enterprises and the GB/T 32151 Requirements on Calculation and Reporting of GHG Emissions published by the Standardization Administration of China.
7. The indirect greenhouse gas emissions for the Shanghai and Wuxi plant were calculated based on the Notice on Adjusting the Emission Factor Values in the Municipal Greenhouse Gas Emissions Accounting Guidelines issued by the Shanghai Municipal Ecology and Environment Bureau and the Announcement on the Release of 2022 CO₂ Emission Factors for Electricity (December 2024) issued by China's Ministry of Ecology and Environment, respectively. In 2024, Shanghai plant used a default value of electricity emission factor of 0.42 tCO₂/MWh, and Wuxi plant used the national average grid emission factor of 0.5366 tCO₂/MWh when calculating.

Employment and Labor Practice

Performance Indicators		Unit	2022	2023	2024
Employment					
Total number of full-time employees		Person	6,760	6,863	7,487
By Gender	Male employees	Person	4,932	5,039	5,525
	Female employees	Person	1,828	1,824	1,962
By Type of Employment	Employees working under a labor contract with the employer	Person	6,760	6,863	7,487
	Employees working under a labor contract with a labor dispatch (not included in the total number of full-time employees)	Person	81	53	49
	Part-time employees	Person	0	0	0
	Employees aged under 30	Person	2,983	2,894	3,228
By Age	Employees aged between 30 and 50	Person	3,624	3,811	4,091
	Employees aged above 50	Person	153	158	168
	Non-management employees	Person	–	–	7,361
By Level	Management employees	Person	–	–	126
	Employees with graduate degree or above	Person	–	–	1,987
By Educational Level	Employees with undergraduate degrees	Person	–	–	2,662
	Employees with less than undergraduate degrees	Person	–	–	2,838
	Employees working in Mainland China	Person	6,751	6,855	7,479
By Region	Employees working in Hong Kong, Macao, Taiwan and overseas	Person	9	8	8
	Employee turnover rate ¹	%	12.0	10.0	11.2
By Gender	Female employee	%	9.7	7.1	6.3
	Male employee	%	12.8	11.3	12.5
By Age	Employees aged under 30	%	18.3	14.0	17.2
	Employees aged between 30 and 50	%	7.2	7.5	6.3
	Employees aged above 50	%	0.7	1.9	1.2
By Region	Employees working in Mainland China	%	12.0	10.0	11.2
	Employees working in Hong Kong, Macao, Taiwan and overseas	%	0.0	9.7	0.0

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Performance Indicators		Unit	2022	2023	2024
Health and Safety ²					
Occupational disease incidence		%	0	0	0
Coverage of employees attending occupational disease medical examinations		%	100	100	100
Number of work-related fatalities		Person	0	0	0
Percentage of work-related fatalities		%	0	0	0
Lost days due to work injury		Day	83	61	126
Employee Training					
Average training hours completed per full-time employee ³		Hour	122.2	133.7	137.3
By Level	Average training hours completed per non-management	Hour	123.7	135.6	139.1
	Average training hours completed per management	Hour	26.5	31.6	29.3
By Gender	Average training hours completed per female	Hour	124.3	125.1	119.6
	Average training hours completed per male	Hour	121.4	136.8	143.6
Percentage of full-time employees trained ⁴		%	100	100	100
By Level	Percentage of non-management employees trained	%	100	100	100
	Percentage of management employees trained	%	100	100	100
By Gender	Percentage of female employees trained	%	100	100	100
	Percentage of male employees trained	%	100	100	100
Total employee training expenditure		RMB ten thousand	—	—	195
Employee Rights					
	Employee labor contract signing rate	%	100	100	100
	Employee social insurance coverage rate	%	100	100	100
	Number of employee discrimination incidents	Case	0	0	0
	Amount of employee work injury insurance input	RMB ten thousand	—	—	963
	Employee work injury insurance coverage rate	%	100	100	100

Notes:

- Employee turnover rate for each category = number of employees in that category turned over during the year/number of employees in that category at the end of the year.
- Coverage rate for employee occupational disease physical examinations is calculated based on employees in occupational risk positions at the end of the Reporting Period. During the Reporting Period, the Company incurred 3 work-related injuries and has strengthened protection education, including safety training and education for all employees of the departments involved, on-site engineering control measures, etc.
- Average training hours for each category of employees = total hours of training received by each category of employees/number of employees in that category at the end of the year.
- Coverage rate of training for each category of employees = number of employees of that category receiving training/number of employees of that category at the end of the year.

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Product Responsibility and Customer Service

Performance Indicators	Unit	2022	2023	2024
Product Responsibility and Customer Service				
Number of incidents of non-compliance with relevant laws and regulations on products and services during the Reporting Period	Case	0	0	0
Product return rate	%	0.11	0.087	0.037
Percentage of products sold subject to recalls for safety and health reasons	%	0	0	0
Customer Service				
Number of complaints received in relation to products and services	Case	0	4	2
Percentage of customer complaints resolved ¹	%	–	100	100

Note:

- 1 Percentage of customer complaints resolved = number of customer complaints resolved/number of complaints received about products and services *100%.

Information security and privacy protection

Indicator Name	Unit	2022	2023	2024
Number of confirmed incidents of leakage, theft or loss of customer information	Case	0	0	0
Amount involved in data security incidents	RMB ten thousand	0	0	0
Amount involved in customer privacy leakage incidents	RMB ten thousand	0	0	0

Supply Chain Management

Performance Indicators	Unit	2022	2023	2024
Total number of suppliers	Supplier	551	561	581
By region				
Total number of suppliers in Mainland China	Supplier	413	420	436
Total number of suppliers from Hong Kong, Macao, Taiwan and overseas	Supplier	138	141	145
Number of suppliers assessed ¹	Supplier	112	114	139
Number of suppliers subject to rectification	Supplier	0	0	0
Percentage of raw and auxiliary material suppliers signing the Environmental Protection Undertaking	%	100	100	100
Percentage of raw materials purchased from local suppliers ²	%	31	32	36

Notes:

- 1 Refers to the number of suppliers for which the Company has conducted assessments in respect of labor, health, environment and ethics.
- 2 Raw material procurement includes: silicon wafers, quartz, target materials, gases, chemicals and other raw materials for production.

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Anti-corruption

Performance Indicators	Unit	2022	2023	2024
Number of concluded legal cases regarding corrupt practices brought against the issuer or its employees during the Reporting Period	Case	0	0	0
Cumulative hours of anti-corruption training received by employees	Hour	/	2,542	2,245
Average training hours per employee on anti-corruption ¹	Hour	/	/	1.2

Note:

- 1 Employees who have received anti-corruption training include all new employees and employees in key positions. Average training hours per employee on anti-corruption = training hours for new employees and employees in key positions/number of new employees and employees in key positions.

Community Investment

Performance Indicators	Unit	2022	2023	2024
Number of employees participating in volunteer services	Person	4,189	1,108	1,100
Cumulative hours of volunteer activities	Hour	6,283	1,364	1,212
Community investment	RMB	–	50,000	50,000

Economic Performance

Performance Indicators	Unit	2022	2023	2024
Social contribution per share ¹	RMB	5.46	3.59	1.87

Note:

- 1 Social contribution per share = (net profit of the Company + tax paid to the state during the year + salaries paid to employees + interest on borrowings paid to banks and other creditors + value created for other stakeholders, e.g. donations – other social costs due to environmental pollution)/total number of shares of the Company.

8.2 List of Laws and Regulations Observed by the Company

Fields	Names of Laws and Regulations
Environmental Responsibility	
Environmental Management	Environmental Protection Law of the People's Republic of China, Environmental Impact Assessment Law of the People's Republic of China, Clean Production Promotion Law of the People's Republic of China, Environmental Protection Tax Law of the People's Republic of China, etc.
Emissions and Waste Management	Law of the People's Republic of China on Prevention and Control of Atmospheric Pollution, Law of the People's Republic of China on the Prevention and Control of Water Pollution, Law of the People's Republic of China on the Prevention and Control of Environmental Pollution by Solid Wastes, Regulations on Administration of Recovery and Disposal of Waste Electrical and Electronic Products, Regulations on the Administration of Pollution Discharge Permits, etc.
Energy and Resource Management	Energy Conservation Law of the People's Republic of China, Law on Energy of the People's Republic of China, Renewable Energy Law of the People's Republic of China, Law on Circular Economy Promotion of the People's Republic of China, Water Conservation Regulations, etc.
Product Quality	Law of the People's Republic of China on Product Quality, Anti-Unfair Competition Law of the People's Republic of China, EU Regulation concerning the Registration, Evaluation, Authorization and Restriction of Chemicals ("REACH"), Waste Electrical and Electronic Equipment Directive ("WEEE"), Restriction of Hazardous Substances ("RoHS"), etc.

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Information Security and Privacy Protection	Cyber Security Law of the People's Republic of China, Data Security Law of the People's Republic of China, Personal Information Protection Law of the People's Republic of China, Cryptography Law of the People's Republic of China, Regulations on the Protection of Computer Software, Regulations on Network Data Security Management
Intellectual Property Protection	Patent Law of the People's Republic of China, Copyright Law of the People's Republic of China, Trademark Law of the People's Republic of China, Implementation Rules of the Patent Law of the People's Republic of China, Regulations on National Science and Technology Awards, etc.
Responsibility for Employees	
Employee Interests	Labor Law of the People's Republic of China, Law of the People's Republic of China on Employment Contracts, Social Insurance Law of the People's Republic of China, Law of the People's Republic of China on Protection of Women's Rights and Interests, Vocational Education Law of the People's Republic of China, Regulations on the Prohibition of Child Labor
Occupational Health and Safety	Law of the People's Republic of China on Prevention and Control of Occupational Diseases, Production Safety Law of the People's Republic of China, Special Equipment Safety Law of the People's Republic of China, Regulation on Work-Related Injury Insurances, Regulations on Labor Protection in Workplaces Using Toxic Substances, Special Regulations on Labor Protection for Female Employees, Regulations on Emergency Response to Production Safety Accidents, etc.
Corporate Governance	
Corporate Governance	Company Law of the People's Republic of China, Securities Law of the People's Republic of China, Hong Kong Companies Ordinance, etc.

8.3 Benchmarking Index

The Hong Kong Stock Exchange's Environmental, Social and Governance Reporting Guide (effective from 31 December 2023)

Part B: Mandatory Disclosure Requirements

Mandatory Disclosure Items	Section in the Report
Governance Structure	ESG Management System
Reporting Principles	Report Preparation Instructions
Reporting Boundary	Report Preparation Instructions

Part C: "Comply or explain" Provisions

Aspects, General Disclosures and KPIs	Section in the Report
Subject Area A. Environmental	
Aspect A1. Emissions	
General Disclosure A1	Emissions and Waste Management
KPI A1.1	Emissions and Waste Management ESG Data and Notes
KPI A1.2	ESG Data and Notes

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Aspects, General Disclosures and KPIs	Section in the Report
KPI A1.3	ESG Data and Notes
KPI A1.4	ESG Data and Notes
KPI A1.5	Emissions and Waste Management ESG Management Strategies and Targets
KPI A1.6	Emissions and Waste Management ESG Management Strategies and Targets
Aspect A2. Use of Resources	
General Disclosure A2	Energy Management Water Resources Management
KPI A2.1	ESG Data and Notes
KPI A2.2	ESG Data and Notes
KPI A2.3	Energy Management ESG Management Strategies and Targets
KPI A2.4	Water Resources Management ESG Management Strategies and Targets
KPI A2.5	ESG Data and Notes
Aspect A3. Environment and Natural Resources	
General Disclosure A3	Environmental Management Water Resources Management Circular Economy
KPI A3.1	Water Resources Management Circular Economy
Aspect A4. Coping with Climate Change	
General Disclosure A4	Climate Change Mitigation and Adaptation
KPI A4.1	Climate Change Mitigation and Adaptation
Subject Area B. Social Employment and Labour Practices	
Aspect B1. Employment	
General Disclosure B1	Rights and Interests of Employees
KPI B1.1	ESG Data and Notes
KPI B1.2	ESG Data and Notes
Aspect B2. Health and Safety	
General Disclosure B2	Employee Health and Safety
KPI B2.1	ESG Data and Notes
KPI B2.2	ESG Data and Notes
KPI B2.3	Employee Health and Safety
Aspect B3. Development and Training	
General Disclosure B3	Employee Development and Training
KPI B3.1	ESG Data and Notes

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Aspects, General Disclosures and KPIs	Section in the Report
KPI B3.2	ESG Data and Notes
Aspect B4. Labor Standards	
General Disclosure B4	Rights and Interests of Employees
KPI B4.1	Rights and Interests of Employees
KPI B4.2	Rights and Interests of Employees
Subject Area B. Social Operating Practices	
Aspect B5. Supply Chain Management	
General Disclosure B5	Sustainable Supply Chain Management
KPI B5.1	ESG Data and Notes
KPI B5.2	Sustainable Supply Chain Management
KPI B5.3	Sustainable Supply Chain Management
KPI B5.4	Sustainable Supply Chain Management
Aspect B6. Product Responsibility	
General Disclosure B6	Product and Service Safety and Quality
KPI B6.1	Product and Service Safety and Quality
KPI B6.2	Product and Service Safety and Quality ESG Data and Notes
KPI B6.3	R&D Innovation and Intellectual Property Protection
KPI B6.4	Product and Service Safety and Quality
KPI B6.5	Information Security and Privacy Protection
Aspect B7. Anti-corruption	
General Disclosure B7	Business Ethics
KPI B7.1	ESG Data and Notes
KPI B7.2	Business Ethics
KPI B7.3	Business Ethics
Aspect B8. Community Investment	
General Disclosure B8	Community and Public Welfare
KPI B8.1	Community and Public Welfare
KPI B8.2	ESG Data and Notes

Self-Regulatory Guidelines No.14 for Companies Listed on Shanghai Stock Exchange—Sustainability Report (For Trial Implementation) (April 2024)

Disclosure Requirement	Corresponding Sections of the Report
Coping with Climate Change	Climate Change Mitigation and Adaptation
Pollutant Emissions	Emissions and Waste Management
Waste Disposal	Emissions and Waste Management

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Disclosure Requirement		Corresponding Sections of the Report
Ecosystem and Biodiversity Conservation		In terms of biodiversity conservation, the four factories of the Company are located in industrial plots and do not involve control requirements such as ecological protection red lines and water source protection areas.
Environmental Compliance Management		Environmental Management
Energy Utilization		Energy Management
Water Resource Utilization		Water Resources Management
Circular Economy		Circular Economy
Rural Revitalization		Community and Public Welfare
Social Contribution		Community and Public Welfare
Innovation-Driven		R&D Innovation and Intellectual Property Protection
Technology Ethics		The Company's core business focuses on the production and manufacturing of semiconductor wafers and does not involve this topic.
Supply Chain Security		Sustainable Supply Chain Management
Equal Treatment for SMEs		As of the end of 2024, the balance of the Company's accounts payable (including notes payable) will not exceed RMB30 billion or account for more than 50% of total assets.
Product and Service Safety and Quality		Product and Service Safety and Quality
Data Security and Customer Privacy Protection		Information Security and Privacy Protection
Employee		Rights and Interests of Employees Employee Health and Safety Employee Development and Training
Due Diligence		Stakeholder Engagement and Due Diligence
Stakeholder Communication		Stakeholder Engagement and Due Diligence
Anti-Commercial Bribery and Anti-Corruption		Business Ethics
Anti-Unfair Competition		Business Ethics
Self-disclosure topics	Green Products	Green Products
	Clean Technology Opportunities	Clean Technology Opportunities
	Risk Compliance Management	Risk Compliance Management
	Corporate Governance	Corporate Governance

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Index Table of Environmental, Social and Governance (ESG) Indicator System for State-owned Listed Companies in Shanghai (Version 1.0)

KPIs		Section in the Report
E Environment		
E1 Environmental Management		
E1.1	Environmental Management Objectives and Systems	ESG Management Strategies and Targets Environmental Management
E1.2	Passing Environmental Management System Certification	Environmental Management
E1.3	Research and Development of Green Products and Use of Environmental Protection Technologies	Green Products Clean Technology Opportunities
E1.4	Total Investment in Environmental Protection	Environmental Management
E1.5	Environmental Protection Training Performance	Environmental Management
E2 Energy		
E2.1	Energy Management Goals and Planning	ESG Management Strategies and Targets Energy Management
E2.2	Energy Saving Measures	Energy Management
E2.3	Renewable Energy Development and Application	Energy Management
E2.4	Total Energy Consumption	ESG Data and Notes
E3 Resources		
E3.1	Resource Management System and Planning	Water Resources Management Circular Economy
E3.2	Water Resource Goals and Planning	Water Resources Management
E3.3	Water Conservation Measures	Water Resources Management
E3.4	Total Annual Water Consumption	ESG Data and Notes
E3.5	Other Resource Usage Management	Green Products
E4 Pollutants		
E4.1	Sewage Management Goals and Planning	ESG Management Strategies and Targets Emissions and Waste Management
E4.2	Measures to Reduce Sewage Discharge	Emissions and Waste Management
E4.3	Sewage Recycling Usage	ESG Data and Notes
E4.4	Sewage Pollutant Discharge	ESG Data and Notes
E4.5	Waste Gas Management Goals and Planning	ESG Management Strategies and Targets Emissions and Waste Management
E4.6	Measures to Reduce Waste Gas Discharge	Emissions and Waste Management
E4.7	Waste Gas Pollutant Emissions	ESG Data and Notes

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KPIs		Section in the Report
E4.8	Solid Waste Management Goals and Planning	ESG Management Strategies and Targets Emissions and Waste Management
E4.9	Solid Waste Treatment Measures	Emissions and Waste Management
E4.10	Solid Waste Emissions	ESG Data and Notes
E4.11	Solid Waste Recycling, Utilization, and Disposal Volume	ESG Data and Notes
E4.12	Other Pollutants Management	The Company adopts noise prevention and control measures such as building noise insulation, installation of vibration-damping devices for equipment, etc.
E5 Climate Change		
E5.1	Identify Climate Change Risks and Establish Goals and Strategies to Address Climate Change	ESG Management Strategies and Targets Climate Change Mitigation and Adaptation
E5.2	Measures to Support the “Emission Peak and Carbon Neutrality” Goal	Climate Change Mitigation and Adaptation
E5.3	Carbon Verification/Review Measures	Climate Change Mitigation and Adaptation
E5.4	Greenhouse Gas Emissions	ESG Data and Notes
E6 Biodiversity		
E6.1	Biodiversity Conservation System	In terms of biodiversity conservation, the four factories of the Company are located in industrial plots and do not involve control requirements such as ecological protection red lines and water source protection areas.
E6.2	Biodiversity Conservation Measures	
S Social		
S1 Products and services		
S1.1	Product Safety and Quality Management System	Product and Service Safety and Quality
S1.2	Passing Product Quality Management System Certification	Product and Service Safety and Quality
S1.3	Product Research and Development Innovation	R&D Innovation and Intellectual Property Protection
S1.4	Intellectual Property Protection	R&D Innovation and Intellectual Property Protection
S1.5	Customer Information Protection and Privacy	Information Security and Privacy Protection
S1.6	Customer Service Management System	Product and Service Safety and Quality
S1.7	Responsible Publicity	Product and Service Safety and Quality
S1.8	Number of Customer Complaints/ Complaint Resolution rate	Product and Service Safety and Quality
S1.9	Customer Satisfaction	Product and Service Safety and Quality

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KPIs		Section in the Report
S2 Employee Responsibility		
S2.1	Staff Employment and Staff Composition	Rights and Interests of Employees
S2.2	Employee Turnover Rate	ESG Data and Notes
S2.3	Democratic Management of Employees	Rights and Interests of Employees
S2.4	Salary and Benefits System	Rights and Interests of Employees
S2.5	Employee Care	Rights and Interests of Employees
S2.6	Employee Satisfaction	Rights and Interests of Employees
S2.7	Employee Occupational Health and Safety Management	Employee Health and Safety
S2.8	Passing Occupational Health and Safety Management System Certification	Employee Health and Safety
S2.9	Safety Emergency Management Measures	Employee Health and Safety
S2.10	Total Investment in Production Safety	Employee Health and Safety
S2.11	Production Safety Training Performance	Employee Health and Safety
S2.12	Work-related Injuries and Deaths	ESG Data and Notes
S2.13	Employee Physical Examination Coverage	ESG Data and Notes
S2.14	Employee Career Development System	Employee Development and Training
S2.15	Employee Training and Practical Measures	Employee Development and Training
S2.16	Employee Development Performance	Employee Development and Training
S3 Supply Chain Responsibility		
S3.1	Supplier Management System	Sustainable Supply Chain Management
S3.2	Supplier ESG Review	Sustainable Supply Chain Management
S4 Community Responsibility		
S4.1	Carrying out Public Welfare and Charity Activities	Community and Public Benefit
S4.2	Public Welfare and Charity Investment	ESG Data and Notes
S4.3	Volunteer Service Performance	ESG Data and Notes
S5 Corporate Responsibility		
S5.1	Serving National Strategy or Shanghai Strategy	R&D Innovation and Intellectual Property Protection Clean Technology Opportunities Community and Public Benefit
S5.2	Responding to Public Crises	Climate Change Mitigation and Adaptation
S5.3	Information Security	Information Security and Privacy Protection
S5.4	Public Services	Community and Public Benefit
S5.5	Tax Contribution	Financial Statements Section of Hua Hong Semiconductor Limited 2024 Annual Report (A Share)

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KPIs		Section in the Report
G Governance		
G1 Corporate Governance		
G1.1	Party Leadership	Environmental, Social Responsibility and Other Corporate Governance Section of Hua Hong Semiconductor Limited 2024 Annual Report (A Share)
G1.2	Separation of Chairman/General Manager	Separation of Chairman and General Manager
G1.3	Percentage of Directors	Corporate Governance
G1.4	Compliance Management System	Risk Compliance Management
G1.5	Risk Management System	Risk Compliance Management
G1.6	Audit System	Business Ethics
G1.7	Measures to Avoid Violating Business Ethics	Business Ethics
G1.8	Anti-monopoly and Fair Competition	Business Ethics
G2 ESG Governance		
G2.1	Board Involvement in ESG Management	Board of Directors ESG Management Statement ESG Management Framework
G2.2	ESG Organizational Structure	ESG Management Framework
G2.3	ESG strategy/objectives	ESG Management Strategies and Targets
G2.4	ESG Stakeholder Identification	Stakeholder Engagement and Due Diligence
G2.5	Identification of Material Topics	Double Materiality Analysis
G2.6	ESG-related System	ESG Management Framework
G2.7	Carrying out Stakeholder Communication Activities	Stakeholder Engagement and Due Diligence
G2.8	Management Remuneration Linked to ESG Performance	Environmental, safety, R&D goals linked to management compensation
G2.9	Specific ESG Factors for Business	Climate Change Mitigation and Adaptation Clean Technology Opportunities
G3 Data Governance		
G3.1	Basic Platform Construction	Employee Health and Safety R&D Innovation and Intellectual Property Protection
G3.2	Data Convergence and Interoperability	Production, environment and procurement data are kept in the Company's intranet, which is physically isolated from the extranet by setting access rights for reasons of commercial confidentiality.
G3.3	Data Governance System	Not Involved



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